

Catalyst BSS — the training course

Business support for a Sovereign: from a lead to a closed month.
Concepts, worked examples and the exact screens.

Environment in the screenshots: hw307 (chargeback.hw307.omani.works), BSS 0.1.53, 27 Sep 2026. Example figures are illustrative unless a slide says they are pinned by a test.

How to use this course

- Twelve modules, each in the same shape: the concept, a worked example with numbers, the screen where it happens, a self-check.
- Follow the journey order the first time; each module builds on the previous one.
- Every screenshot is the live console on hw307. The page path is written under the title so you can open the same screen while you read.
- Money examples use OMR with 5 % tax. Where a figure is pinned by an automated test, the slide says so — those are exact.
- Terms: a Sovereign runs Catalyst; it bills Customers; a customer's usage arrives through Sources; a Blueprint is not a BSS concept.

The journey — one line, twelve modules

Step	What happens	Module	Console page
1	A prospect prices a basket and leaves an address	Leads	/estimate → /leads
2	The customer is created with its commercial profile and its sources	Customers & sources	/customers
3	Collectors measure what the sources use	Metering	/resources
4	Usage is priced: price books, plans, discounts, tax	Price books & rating	/pricebooks · /discounts · /tax
5	Negotiated terms: allowances, commitments, minimums, SLA	Contracts	/contracts
6	Everyone reads the cost: overview, explorer, budgets, anomalies	Cost analysis	/overview · /explore
7	The period is run, issued, sent, paid, chased	Billing	/statements · /collections · /billing
8	Money is posted, reconciled and the month is closed	Finance	/finance/*
9	Selling through resellers and agents	Partners	/partners
10	What the Sovereign can sell: pools, shapes, placements	Capacity	/capacity
11	Who may do what, and what the product sends	Access & notifications	/access · /notifications

Two layers that never meet in billing

Cloud layer — resold National Cloud

- A customer's cloud project (ECS, EVS, EIP, RDS...) collected read-only through AK/SK.
- Priced by a cloud-scoped book, e.g. 'National Cloud 2026 list'.
- The Sovereign's own cloud project is the 'landlord': its cost is the pool that Allocation spreads — as a report, never a bill.

Platform layer — Organizations on Catalyst

- An Organization on this Sovereign, collected by the platform collector (plan-hours, k8s vCPU, memory, PVC).
- Priced by a platform-scoped book: 'OpenOva plans' (committed) or 'Organization PAYG' (pay-per-use).
- The per-Organization platform stack is excluded from the customer's usage and shown as Platform overhead.

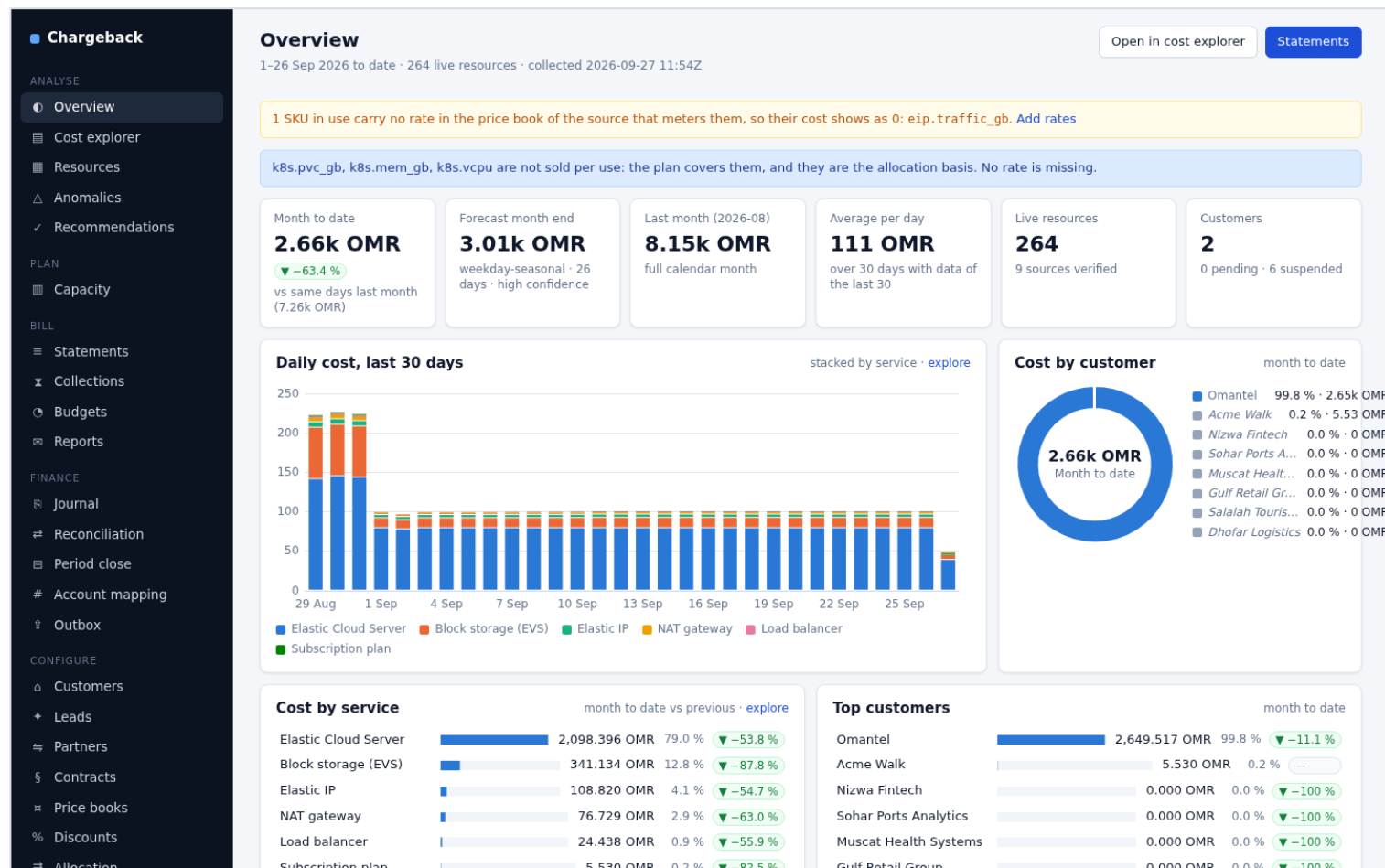
A source belongs to exactly one layer and is priced by exactly one book. That is the whole ownership model.

Who does what — the six roles

Role	Scope	Can
sovereign-admin	Sovereign	everything, including access and settings
billing-operator	Sovereign	read, price books, customers, issue, collect, audit, capacity
finance-viewer	Sovereign	read and export only, plus audit
customer-owner	one customer	read its own cost, top up its account, manage its users, sources, PO and tax registration
customer-billing	one customer	read and top up
customer-viewer	one customer	read

A partner principal gets a third scope, partner:<id>, that expands to the customers assigned to it (module 9).

The console, oriented



1. Left navigation: the modules in journey order — Customers, Resources, Price books, Contracts, Statements, Collections, Finance, Partners, Capacity, Access, Notifications.

2. The Overview is the Sovereign-wide cost picture: month to date, compared with the previous window.

3. Everything on this page is scoped by your role: a customer-owner sees only its own customer here (as /my/overview).

01

Leads and the public calculator

How a prospect prices a basket before anyone has created a customer — and what becomes a lead.

In this module: What is public and what never is · One pricing function, two callers · Estimate → lead → customer · The calculator, screen by screen

What is public — and what never is

- Public: one list price book the Sovereign chose to publish, the catalogue plans, and a pricing endpoint that prices a basket from them.
- Never public: customers, discounts, contracts, usage, statements. An estimate belongs to nobody — no customer, no session, no cookie.
- One pricing function: the public estimate and the billing run call the same rating code on the same book, so an estimate never quotes a price the bill could not produce.
- A saved estimate is valid 30 days and has a shareable link.

Estimate, lead, customer — three different things

- Estimate: a priced basket. Saved with its lines, currency, region, monthly and yearly totals, and the name of the book it was priced from.
- Lead: an estimate on which the prospect left an email. Nothing else changes; the flag makes it appear on the Leads page.
- Leads page (/leads): a read-only list — received, address, per month, per year, the link. It needs customers.manage.
- Customer: created by the operator in Customers (module 2). The lead is context for that conversation; nothing converts by itself.

The public calculator — a catalogue by product family

Cost calculator

Choose the services you need and see what a month costs, at list prices. Tax is shown separately, and nothing here needs an account. Prices as of 2026-09-11.

Region

Any region

The region applies to every item. Prices are the same in every region of this list.

Find a service

Search by name — servers, database, backup...

Compute

Servers, and the scaling that keeps up with demand.

2 services

Elastic Cloud Server

Virtual servers — general purpose, compute-optimised or memory-optimised, with a disk and an Elastic IP if you want them.

from 11.676 OMR / month

Configure

Auto Scaling

Adds and removes servers with demand.

from 0.000 OMR / month

Configure

Storage

Disks, backups and images, priced per GB.

3 services

Block storage

Disks for your servers — SSD or HDD, priced per GB.

from 0.035 OMR / month per GB

Configure

Backup

Backups of servers and disks, priced per GB kept.

from 0.049 OMR / month per GB

Configure

Images

Private server images, priced per GB kept.

from 0.000 OMR / month per GB

Configure

Networking

Addresses, load balancing, gateways and protection.

7 services

Elastic IP

Your estimate

Per month

12 months

incl. tax

incl. tax

Choose a service on the left to start an estimate

Each service opens a short form. Every figure comes back priced as an invoice would be.

Subtotal

—

Tax (5.0 %)

—

Share estimate

Send me this estimate

you@company.com

Send

Priced from **National Cloud 2026 list** · prices as of 2026-09-11.
List prices. Taxes are shown separately. A negotiated price, a discount or a partner rate is never part of this estimate; contact us for a proposal.

1. Region at the top applies to every line; 'Find a service' searches by plain name.

2. Families — Compute, Storage, Networking, Databases, Containers, Platform plans — each service with a one-line description and its 'from' price.

3. The estimate on the right starts empty and tells the prospect what to do; 'Send me this estimate' with an address is what makes a lead.

Configuring a service — no SKU typed anywhere

Cost calculator
Choose the services you need and see what a month costs at list prices. Taxes shown separately and nothing here needs an account. Prices as of 2025-08-11.

Region
Any region

Find a service
Search by name

Compute
Servers, and more

Elastic Cloud Server
from 11.676 OMR / month

Auto Scaling
from 0.000 OMR / month

Storage
Disks, backups and more

Block storage
from 0.035 OMR / month

Backup
from 0.049 OMR / month

Images
from 0.000 OMR / month per GB. Private server images, priced per GB kept.

Networking
Addresses, load balancing, gateways and protection.

Elastic IP

Elastic Cloud Server
Virtual servers — general purpose, compute-optimised or memory-optimised, with a disk and an Elastic IP if you want them.

Family
Compute-optimised

Servers
1

Usage
Always on (730 h/month)

Size

2 vCPU · 4 GB 35.577 OMR / month	2 vCPU · 8 GB 44.474 OMR / month	4 vCPU · 8 GB 71.160 OMR / month	4 vCPU · 16 GB 88.949 OMR / month	8 vCPU · 16 GB 142.314 OMR / month
8 vCPU · 32 GB 177.897 OMR / month	12 vCPU · 24 GB 213.474 OMR / month	12 vCPU · 48 GB 266.846 OMR / month	16 vCPU · 32 GB 284.635 OMR / month	16 vCPU · 64 GB 355.788 OMR / month
24 vCPU · 48 GB 426.949 OMR / month	24 vCPU · 96 GB 533.686 OMR / month	32 vCPU · 64 GB 569.263 OMR / month	32 vCPU · 128 GB 711.583 OMR / month	48 vCPU · 96 GB 853.897 OMR / month
48 vCPU · 192 GB 1,067.372 OMR / month	64 vCPU · 128 GB 1,138.532 OMR / month	64 vCPU · 256 GB 1,423.160 OMR / month	96 vCPU · 192 GB 1,707.795 OMR / month	96 vCPU · 384 GB 2,134.744 OMR / month

Attached disk
No disk
Disks are kept for the whole month

☐ Add an Elastic IP to each server

This adds 1 line to the estimate
Compute-optimised · 2 vCPU · 4 GB ecs.c7n.large.2 1 × 730 h

Cancel Add to estimate

Configure

List prices. Taxes are shown separately. A negotiated price, a discount or a partner rate is never part of this estimate; contact us for a proposal.

1. Family (general purpose / compute-optimised / memory-optimised), number of servers, usage (always on 730 h · business hours 176 h · custom).

2. Size is a grid of chips: vCPU · GB with the monthly price; the SKU resolves from the choice and shows only as a small hint.

3. An attached disk and an Elastic IP per server become extra lines under the same item.

The estimate — grouped, visual, editable

Cost calculator
Choose the services you need and see what a month costs, at list prices. Tax is shown separately, and nothing here needs an account. Prices as of 2026-09-11.

Region
Any region The region applies to every item. Prices are the same in every region of this list.

Find a service
Search by name — servers, database, backup...

Compute 2 services
Servers, and the scaling that keeps up with demand.

Elastic Cloud Server
Virtual servers — general purpose, compute-optimised or memory-optimised, with a disk and an Elastic IP if you want them. Configure

Auto Scaling
Adds and removes servers with demand. Configure

Storage 3 services
Disks, backups and images, priced per GB.

Block storage
from 0.035 OMR / month per GB Disks for your servers — SSD or HDD, priced per GB. Configure

Backup
from 0.049 OMR / month per GB Backups of servers and disks, priced per GB kept. Configure

Images
from 0.000 OMR / month per GB Private server images, priced per GB kept. Configure

Networking 7 services
Addresses, load balancing, gateways and protection.

Elastic IP Configure

Your estimate 2 items · 2 lines · OMR

Per month **115.770** OMR incl. tax

12 months **1,389.234** OMR incl. tax

97 % Compute is the largest share of 2 families

■ Compute 106.731 OMR 97 %
■ Storage 3.526 OMR 3 %

COMPUTE > ELASTIC CLOUD SERVER

Elastic Cloud Server
3 × Compute-optimised · 2 vCPU · 4 GB · always on
106.731 OMR
per month · before tax
Hide lines Edit Remove
Compute-optimised · 2 vCPU · 4 GB ecs.c7n.large.2 · 3 × 730 h
106.731 OMR

STORAGE > BLOCK STORAGE

Block storage **3.526 OMR**
1 × 100 GB HDD per month · before tax
Hide lines Edit Remove
HDD · 100 GB each evs.hdd.gb · 100 × 730 h
3.526 OMR

Subtotal **110.257 OMR**

1. Per month and 12 months in large type, tax included; the gauge and the bar show each family's share.

2. Items grouped by family > service with their lines; Edit re-opens the configurator with its values, Remove drops the item.

3. Every figure is priced by the server from the published list book — the same function that prices an invoice.

Leads — what the prospect left behind

■ Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Leads

1

1 still within their 30-day validity

Priced per month

1.57k OMR

the sum of every lead's monthly figure, at list price

RECEIVED ↓	CONTACT ↑	PER MONTH ↑	PER YEAR ↑	WHAT THEY PRICED ↑	REGION ↑	
2026-09-27 11:14Z valid until 2026-10-27	emrahbaysal@gmail.com	1,569,230 OMR	18,830,757 OMR	4 lines L plan · ecs.c7n.6xlarge.4 · cbr.gb	me-east-215	Open estimate

1 row

1. One row per estimate that carries an address, newest first.
2. Per month and per year are the totals the prospect saw, in the book's currency.
3. The link reopens the estimate exactly as saved; use it in the first conversation, then create the customer.

Leads — check yourself

- Q1. A prospect prices 3 servers and closes the tab without an email. Is that a lead?
- Q2. Can a public estimate ever include a customer's negotiated discount?
- Q3. Where is the price an estimate uses defined?

Answers

- A1. No. It is a saved estimate; only an estimate with an address is a lead.
- A2. No. Discounts are per customer and the estimate has none; it prices from the one published list book.
- A3. In the published price book (module 4), read through the same rating function billing uses.

02

Customers and sources

The billed party, its commercial profile, and where its usage comes from.

In this module: Customer = party · The four commercial fields · Sources and their price book · Users, invites, import
· The customer page, tab by tab

A customer is the party that is billed

- One row per party: name, currency, PO reference, payment terms, tax profile, contacts.
- A partner is also a party and owns a customer row flagged partner (module 9) — one ledger for everyone who owes or is owed money.
- Everything downstream is scoped by customer: usage, cost, statements, account, budgets, users, audit.
- Party kinds you will meet on hw307: Gulf Retail Group, Nizwa Fintech, Dhofar Logistics, Muscat Health Systems, Omantel, Acme Walk (customers) and Walk Reseller (partner).

The four commercial fields decide the whole billing path

Field	Values	Meaning
charging	billed · informational	billed produces invoices; informational is showback only
payment_model	prepaid · postpaid	prepaid settles an invoice from the account balance at issue and service depends on that balance; postpaid leaves it open on terms
payment_method	transfer · gateway · internal	how money arrives: bank transfer against the invoice, a card gateway, or an internal recharge with no cash
gateway	omantel · stripe · —	which gateway collects, when the method is gateway

Profiles you will actually configure

Who	charging	payment_model	method	gateway	Reads as
Omantel corporate, invoiced against a PO on terms	billed	postpaid	transfer	—	real
Omantel SME on Omantel's gateway	billed	prepaid	gateway	omantel	real
The same SME today, on Stripe	billed	prepaid	gateway	stripe	real
An internal department recharged	billed	postpaid	internal	—	chargeback
A department that only wants to see its cost	informational	—	—	—	showback

Creating a customer

The screenshot shows the 'New customer' form in the Catalyst BSS interface. The form is a modal dialog with the following fields and options:

- Name:** Acme Trading LLC
- Slug:** acme (Note: Lowercase; used in statement ids and invite links. Cannot change later.)
- Admin email:** finance@example.com (Note: Receives the invite and signs in as customer-admin.)
- Kind:** External (Note: An external account billed for its own cloud projects.)
- Charging:** Informational (Note: Statements exist for visibility only; nothing is ever collected.)
- Start date:** mm/dd/yyyy (Note: Usage before this day is not billed. Empty = from the first collection.)

At the bottom of the form are 'Cancel' and 'Create and add a source' buttons. Below the form is a table of existing customers:

CUSTOMER	Status	Kind	Charging	Balance	Due	Last Collected	Last Statement	Action
Acme Walk acmewalk307 · organization emrah.baysal@openova.io	Active	External	Informational	0.000 OMR	20.450 OMR in credit	2026-09-27 11:54Z	2026-09	Re-invite
Dhofar Logistics demo-dhofar-logistics · accounts@dhofarlogistics.example	Suspended	Internal recharge	net 30	0.000 OMR	71.250 OMR owes	2026-09-01 00:00Z	2026-08	Re-invite
Gulf Retail Group demo-gulf-retail · billing@gulftail.example	Active	Informational	2 cloud 2/2 verified	2,649.517 OMR	5,725.127 OMR owes	2026-09-27 11:46Z	2026-09	Re-invite
Muscat Health Systems demo-muscat-health · finance@muscathealth.example	Suspended	Internal recharge	net 30	0.000 OMR	28.117 OMR	2026-09-01 00:00Z	2026-08	Re-invite
Nizwa Fintech demo-nizwa-fintech · organization · ops@nizwafintech.example	Suspended	Internal recharge	net 30	0.000 OMR	28.117 OMR	2026-09-01 00:00Z	2026-08	Re-invite
Omantel hw307-omani-works · emrah.baysal@openova.io	Active	Informational	2 cloud 2/2 verified	2,649.517 OMR	5,725.127 OMR owes	2026-09-27 11:46Z	2026-09	Re-invite
Salalah Tourism Board demo-salalah-tourism · organization · digital@salalah-tourism.example	Suspended	Internal recharge	net 30	0.000 OMR	71.250 OMR owes	2026-09-01 00:00Z	2026-08	Re-invite
Sohar Ports Analytics demo-sohar-ports ·	Suspended	Internal recharge	net 30	0.000 OMR	28.117 OMR	2026-09-01 00:00Z	2026-08	Re-invite

1. Name, currency and the commercial profile are set here; the tax profile and PO can follow on the Settings tab.

2. Nothing is billed until a source is attached and a period is run.

3. Bulk onboarding: /customers/import takes a CSV with the same fields.

The customer list

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Customers

8 customers · 2 active · 0 pending · 6 suspended

Import CSV

New customer

Active

2

collected and billed

Pending

0

every customer is activated

Suspended

6

not collected, sign-in refused

Month to date, all customers

2.66k OMR

▼ -63.4 %

vs same days last month

Search

name, slug or admin email

Status

All

Active

Pending

Suspended

CUSTOMER ↑	STATUS ↑	CHARGING ↑	SOURCES ↑	MONTH TO DATE ↑	BALANCE ↑	LAST COLLECTED ↑	LAST STATEMENT ↑	
Acme Walk acmewalk307 · organization · emrah.baysal@openova.io	active	prepaid · Stripe	1 platform 1/1 verified	5.530 OMR	settled	2026-09-27 11:54Z	2026-09	Re-invite
Dhofar Logistics demo-dhofar-logistics · accounts@dhofarlogistics.example	suspended	prepaid · Stripe	1 cloud 1/1 verified	0.000 OMR	3,821.244 OMR owes	2026-08-25 00:00Z	2026-08	Re-invite
Gulf Retail Group demo-gulf-retail · billing@gulfretail.example	suspended	prepaid · Stripe	1 cloud 1/1 verified	0.000 OMR	5,164.047 OMR owes	2026-08-31 23:00Z	2026-08	Re-invite
Muscat Health Systems demo-muscat-health · finance@muscathealth.example	suspended	prepaid · Stripe	1 cloud 1/1 verified	0.000 OMR	6,244.825 OMR owes	2026-09-01 00:00Z	2026-08	Re-invite
Nizwa Fintech demo-nizwa-fintech · organization · ops@nizwafintech.example	suspended	internal recharge net 30	1 platform 1/1 verified	0.000 OMR	20.450 OMR in credit	2026-09-01 00:00Z	2026-09	Re-invite
Omantel hw307-omani-works · emrah.baysal@openova.io	active	informational	2 cloud 2/2 verified	2,649.517 OMR	5,725.127 OMR owes	2026-09-27 11:46Z	2026-09	Re-invite
Salalah Tourism Board demo-salalah-tourism · organization · digital@salalah tourism.example	suspended	internal recharge net 30	1 platform 1/1 verified	0.000 OMR	71.250 OMR owes	2026-09-01 00:00Z	2026-08	Re-invite
Sohar Ports Analytics demo-sohar-ports ·	suspended	internal recharge	1 platform	0.000 OMR	28.117 OMR	2026-09-01 00:00Z	2026-08	Re-invite

- 1. One row per party with its commercial profile and month-to-date cost.
- 2. Open a customer to get its twelve tabs — the rest of this module walks them.

Sources — where a customer's usage comes from

- Cloud source: a National Cloud project, read-only AK/SK, a region, an optional scope token. The cloud collector measures it.
- Platform source: an Organization on this Sovereign. The platform collector measures its plan-hours and Kubernetes meters.
- Each source is assigned ONE price book. The book decides the layer's prices; the customer's discounts and contract apply on top.
- Disable a source to stop collecting without deleting history. A source's credentials belong to the customer-owner to rotate.

Sources tab — attach, assign a book, disable

Chargeback

ANALYSE

- Overview
- Cost explorer
- Resources
- Anomalies
- Recommendations

PLAN

- Capacity

BILL

- Statements
- Collections
- Budgets
- Reports

FINANCE

- Journal
- Reconciliation
- Period close
- Account mapping
- Outbox

CONFIGURE

- Customers
- Leads
- Partners
- Contracts
- Price books
- Discounts
- Allocation

Customers / Gulf Retail Group

Gulf Retail Group suspended

demo-gulf-retail · external · prepaid · Stripe · 1 cloud source · from 2026-06-01 · billing@gulfretail.example

Re-invite Resume Edit Delete

Month to date
0 OMR
▼ -100 %
vs same days last month (1.49k OMR)

Forecast month end
—
needs one complete day

Last month (2026-08)
1.71k OMR
full calendar month

Live resources
0
1 verified source

Statements
3
0 draft · 3 issued

Overview Account Cost Cost centres Resources Statements **3** Contract Discounts Budgets **Sources 1** Users **1** Settings Audit

1 source · 1 verified Add source

KIND	LAYER	REGION	PROJECT	PRICE BOOK	SCOPE	STATUS	ACCESS KEY	VERIFIED	LAST COLLECTED	LAST ERROR	
File import	Cloud	me-east-215-a	demo-gulf-retail	National CM	whole project	verified	none	2026-06-01 00:00Z	2026-08-31 23:00Z	—	<a>Edit <a>Verify <a>Disable <a>Rotate key <a>Delete

1 row

A new source starts pending: add its credential with Rotate key, then Verify. A verified source is collected hourly while the customer is active. **Scope**: Bills only resources whose name carries this token (e.g. a deployment id) — empty bills the whole project. **Price book**: The rate card that prices THIS source. A cloud source takes a cloud book, a platform source a platform book — the two layers are never priced by the same card. **Disable**: a decommissioned source collects nothing new and counts as neither verified nor live, but its collected history still rates — in the explorer and on every statement already issued from it.

1. Add a cloud project (AK/SK, region) or link an Organization.

2. Price book per source: this is where 'which prices apply' is decided.

3. The last-collected stamp tells you the collector is alive for this source.

Settings tab — the commercial profile

Customers / Gulf Retail Group Re-invite Resume Edit Delete

Gulf Retail Group suspended

demo-gulf-retail · external · prepaid · Stripe · 1 cloud source · from 2026-06-01 · billing@gulfretail.example

Month to date
0 OMR
vs same days last month (1.49k OMR)

Forecast month end
—
needs one complete day

Last month (2026-08)
1.71k OMR
full calendar month

Live resources
0
1 verified source

Statements
3
0 draft · 3 issued

Overview Account Cost Cost centres Resources Statements **3** Contract Discounts Budgets Sources **1** Users **1** **Settings** Audit

Customer settings

Name: Gulf Retail Group Slug: demo-gulf-retail
Fixed at creation — it is in every statement id and invite link.

Admin email: billing@gulfretail.example Status: suspended
Signs in as customer-admin and receives invites. Not collected; sign-in refused.

How this customer is charged

Charging: Billed Start date: 06/01/2026
Statements are invoices and are collected. Usage before this day is not billed.

Payment model: Prepaid Payment method: Payment gateway
Pays ahead, or holds a balance that is debited when a statement is issued. A payment gateway collects; choose which one below.

Payment gateway: Stripe
Which gateway collects. Adding another is a deployment change, not a customer one.

Account credit

☐ Auto-apply credit to new invoices

1. charging / payment model / method / gateway — the four fields from the table.

2. PO reference and payment terms are copied onto every draft statement at run time.

3. Tax registration and exemption feed the tax rules (module 4).

Users tab — who signs in for this customer

The screenshot shows the 'Users' tab for the 'Gulf Retail Group' customer. The interface includes a sidebar with navigation options like 'Chargeback', 'ANALYSE', 'PLAN', 'BILL', 'FINANCE', and 'CONFIGURE'. The main content area displays customer details, a summary of key metrics (Month to date OMR, Forecast month end, Last month OMR, Live resources, Statements), a tabbed interface with 'Users' selected, and a table of users. Below the table is a form to add a new user.

Customers / Gulf Retail Group suspended Re-invite Resume Edit Delete

demo-gulf-retail · external · prepaid · Stripe · 1 cloud source · from 2026-06-01 · billing@gulfretail.example

Month to date
0 OMR
vs same days last month (1.49k OMR)

Forecast month end
—
needs one complete day

Last month (2026-08)
1.71k OMR
full calendar month

Live resources
0
1 verified source

Statements
3
0 draft · 3 issued

Overview Account Cost Cost centres Resources Statements **3** Contract Discounts Budgets Sources **1** **Users 1** Settings Audit

The admin email **billing@gulfretail.example** is always an owner. Users sign in at the Sovereign SSO or with a one-time PIN mailed to them; an **owner** reads, tops up the account and manages users, PO reference and tax registration; **billing** reads and tops up; a **viewer** only reads. Recording a payment, issuing, credit notes and suspension stay with the operator.

EMAIL ↕	ROLE ↕	MAY ↕	
billing@gulfretail.example	Owner	reads, tops up, manages users, PO reference and tax registration	Remove

1 row

Email
name@example.com

Role
Viewer — reads costs, resources and invoices

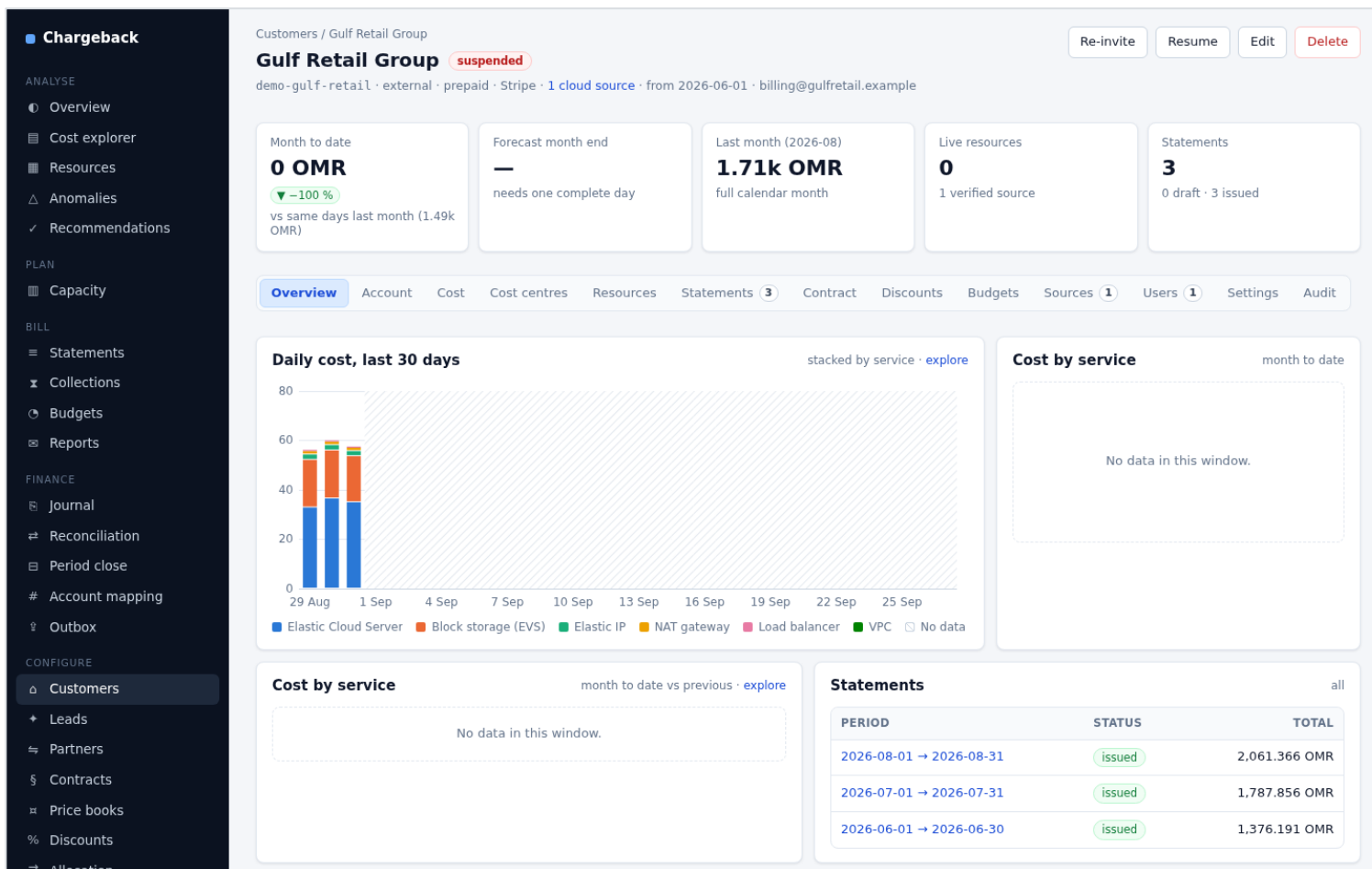
Add user

1. Invite sends an activation link (notification customer.invite); the user gets a customer-scoped role.

2. customer-owner may manage users, sources and the PO; customer-viewer only reads.

3. The same people see the customer lens at /my/* — the pages you see here, confined to their customer.

Overview tab — the customer's month



1. Month to date, previous window, unpriced and not-sold-per-use notes, top services.

2. This is the page that answered in 10 s before 0.1.52 and answers from one ledger read since 0.1.53.

Audit tab — every change to this customer

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Customers / Gulf Retail Group

Gulf Retail Group

suspended

demo-gulf-retail · external · prepaid · Stripe · 1 cloud source · from 2026-06-01 · billing@gulfretail.example

Re-invite

Resume

Edit

Delete

Month to date

0 OMR

vs same days last month (1.49k OMR)

Forecast month end

—

needs one complete day

Last month (2026-08)

1.71k OMR

full calendar month

Live resources

0

1 verified source

Statements

3

0 draft · 3 issued

Overview

Account

Cost

Cost centres

Resources

Statements 3

Contract

Discounts

Budgets

Sources 1

Users 1

Settings

Audit

AT ↓	ACTOR ↑	ACTION ↑	DETAILS
2026-09-27 08:15Z	emrah.baysal@openova.io	account.refund	{"amount":1,"reason":"walk 0.151: refund of the walk top-up","reference":"WALK-0151","payment_id":6}
2026-09-27 08:15Z	emrah.baysal@openova.io	account.allocate	{"auto":false,"allocated":1,"payment_id":6,"allocations":[{"id":7,"amount":1,"payment_id":6,"allocated_at":"2026-09-27T08:15:14.593004Z","allocated_by":"emrah.baysal@openova.io","statement_id":"644e3641-86f8-4893-8d94-41e94098ab79","invoice_number":"INV-2026-00002"}],"unallocated":0}
2026-09-27 08:15Z	emrah.baysal@openova.io	account.payment	{"amount":1,"method":"transfer","gateway":"billing","paid_at":"2026-09-27T08:00:00Z","purpose":"checkout","allocated":0,"reference":"WALK-0151","payment_id":6,"unallocated":1,"payment_status":"received"}
2026-09-14 14:34Z	system	collections.reminder	{"stage":14,"recipients":["billing@gulfretail.example"],"outstanding":1787.856125,"statement_id":"d62ff59c-49f9-458a-94da-b5bd8680f9b2","days_past_due":14,"invoice_number":"INV-2026-00012"}
2026-09-14 14:33Z	system	collections.escalation	{"action":"notify","recipients":["billing@gulfretail.example"],"outstanding":1376.190872,"statement_id":"644e3641-86f8-4893-8d94-41e94098ab79","days_past_due":45,"invoice_number":"INV-2026-00002"}
2026-09-10 12:53Z	emrah.baysal@openova.io	statement.credit_note	{"kind":"partial","total":61.366,"number":"CN-2026-00001","reason":"Goodwill credit - demo walk 2026-09-10","applied":61.366,"unapplied":0,"statement_id":"4cdf26c-3a02-4bfff-aaeb-7ae43096aed2","credit_note_id":"af9e3126-a65f-49f1-a6d7-891a90fe97f0","invoice_number":"INV-2026-00014"}
2026-09-10 12:17Z	system	collections.reminder	{"stage":7,"recipients":["billing@gulfretail.example"],"outstanding":1787.856125,"statement_id":"d62ff59c-49f9-458a-94da-b5bd8680f9b2","days_past_due":10,"invoice_number":"INV-2026-00012"}
			{"stage":0,"recipients":["billing@gulfretail.example"],"outstanding":1787.856125,"s

1. Who changed what and when: profile edits, source changes, invitations, suspensions.

2. audit.read is a Sovereign permission; a customer does not read its own trail.

Customers — check yourself

- Q1. Which field makes an invoice settle from the account at issue?
- Q2. A customer has a cloud project and an Organization. How many price books apply?
- Q3. Who may rotate a source's AK/SK: the operator, the customer-owner, or both?

Answers

- A1. `payment_model = prepaid`.
- A2. Two — one per source: a cloud-scoped book for the project, a platform-scoped book for the Organization.
- A3. Both: `customers.manage` at the Sovereign, `customer.self.manage` for the owner on its own customer.

03

Metering

What the collectors measure, how a record looks, and the two traps: resizes and Elastic IPs.

In this module: Two collectors · Resources and usage records · A resize splits at the minute · The Elastic IP meters by its charge mode · Platform overhead

Two collectors, one record shape

	Cloud collector	Platform collector
Reads	National Cloud project through AK/SK	the Sovereign's own cluster
Inventory	every 15 minutes: ECS, EVS, EIP, ELB, NAT, RDS, DDS, GaussDB, CCE...	Organizations and their pods, PVCs
Events	CTS change events (create, delete, resize)	controller events
Metrics	CES hourly (CPU utilisation, EIP traffic)	requests and usage per pod
Writes	resources (inventory + tags) and usage_records (sku, unit, quantity, window)	the same two tables
Meters	ecs.*, evs.*, eip, eip.bandwidth_mbps, eip.traffic_gb, rds.*, ...	plan.* (plan-hours), k8s.vcpu, k8s.mem_gb, k8s.pvc_gb

A usage record, read aloud

```
usage_records
source_id      1169e528-...    (Gulf Retail Group · cloud project)
resource_id    ecs-0a4f...      (server 'erp-app-01')
sku            ecs.m7n.2xlarge.8    unit instance-hour
quantity       1.000000
window         2026-09-27 10:00 → 11:00 UTC
labels.tags    {env: prod, cost-centre: erp}
```

One row per resource per hour per meter.
The rating run multiplies quantity × the source's book price
and writes the priced ledger (module 4).

- Quantity is time-weighted: a server that existed 30 minutes of the hour has quantity 0.5.
- Tags travel with the record — cost centres (module 6) and the explorer's tag grouping read them here.
- The SKU is the Huawei flavour name: m7n = memory-optimised, 2xlarge = 8 vCPU, .8 = 8 GB per vCPU → 64 GB.

A resize is billed at both sizes, split at the minute

```
10:00 ecs.c7n.xlarge.2 (4 vCPU · 8 GB) running
10:36 resize → ecs.c7n.2xlarge.2 (8 vCPU · 16 GB)
```

records for the 10:00–11:00 window

ecs.c7n.xlarge.2	quantity 0.600000	(36 min)
ecs.c7n.2xlarge.2	quantity 0.400000	(24 min)

a workload that never resizes emits byte-identical records to before – pinned by test.

- Before this fix an in-place resize was billed at whichever size the collector happened to see. Now the CTS event marks the minute.
- The same rule holds on the platform layer for a pod resized in place.

The Elastic IP meters by its charge mode

Address shape	Hourly address fee	Reservation	Traffic
bandwidth-billed (a reserved pipe)	yes	$\text{eip.bandwidth_mbps} \times \text{Mbps} \times \text{hours}$	—
traffic-billed (pay per GB)	yes	—	eip.traffic_gb per GB, hourly from CES
shared bandwidth (several addresses on one pipe)	yes	once per pipe, not per address	—

hw307's own Elastic IPs are traffic-billed: the collector stores `bandwidth_charge_mode` per address and never bills a reservation the cloud did not make.

Platform overhead is measured, not billed to the customer

- Each Organization carries a platform stack the customer did not ask for: Keycloak, newapi, openclaw, Agenity, the vCluster control plane.
- The platform collector excludes those pods from the customer's usage and books them on the Platform overhead line.
- Allocation (module 6) shows overhead beside each customer's share, so the landlord cost is explained, not hidden.
- The host namespace quota is the purchased plan plus that stack — derived from the rendered charts and pinned by tests.

Resources — the inventory with its cost

Chargeback

ANALYSE

- Overview
- Cost explorer
- Resources
- Anomalies
- Recommendations

PLAN

- Capacity

BILL

- Statements
- Collections
- Budgets
- Reports

FINANCE

- Journal
- Reconciliation
- Period close
- Account mapping
- Outbox

CONFIGURE

- Customers
- Leads
- Partners
- Contracts
- Price books
- Discounts
- Allocation

Resources

29 Aug - 27 Sep 2026 · 418 resources · costs at list rates

Export CSV

Period: Last 30 days
From: 08/29/2026
To: 09/27/2026
Kind: All kinds
Region: All regions
Status: All Live Stopped Deleted

Search: name or resource id

Resources: 418
seen in 29 Aug - 27 Sep 2026

Cost in window: 3.32k OMR
29 Aug - 27 Sep 2026 · all 418 rows

Live: ...
present at the last collection

Stopped: 0
billed as compute while the price book says so

Deleted: ...
gone; billed up to the deletion

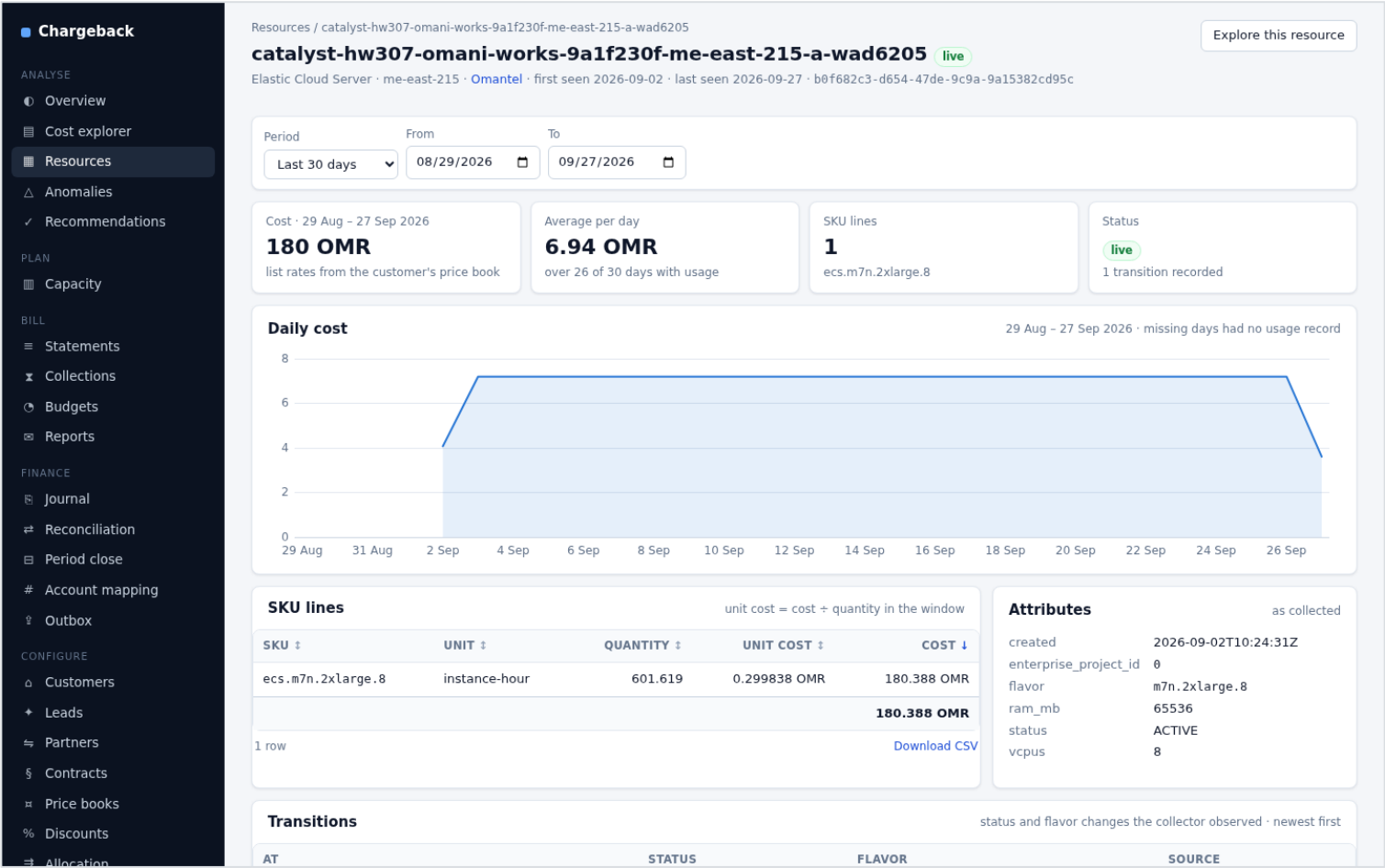
NAME ↕	KIND ↕	CUSTOMER	REGION	STATUS	COST · 29 AUG - 27 SEP 2026 ↓	FIRST SEEN ↕	LAST SEEN ↕
catalyst-hw307-omani-works-9a1f230f-me-east-215-a-wad6205b0f682c3-d654-47de-9c9a-9a15382cd95c	Elastic Cloud Server	Omantel	me-east-215	live	180.313 OMR 1 SKU	2026-09-02	2026-09-27
catalyst-hw307-omani-works-9a1f230f-me-east-215-b-w9e1dc9cea3ecb7-e41a-47c4-b13e-a8d40f629238	Elastic Cloud Server	Omantel	me-east-215	live	180.313 OMR 1 SKU	2026-09-02	2026-09-27
catalyst-hw307-omani-works-9a1f230f-me-east-215-a-we768dc1fc68d10-9d30-45a0-a0c4-320e0b904e90	Elastic Cloud Server	Omantel	me-east-215	live	180.309 OMR 1 SKU	2026-09-02	2026-09-27
catalyst-hw307-omani-works-9a1f230f-me-east-215-a-w9b478732aa7cf8-28ad-41ff-a7c8-d2d1423af60c	Elastic Cloud Server	Omantel	me-east-215	live	180.309 OMR 1 SKU	2026-09-02	2026-09-27
catalyst-hw307-omani-works-9a1f230f-me-east-215-a-w87516b9dadbe23-b6d4-48e6-afc7-fb4bdd64d49	Elastic Cloud Server	Omantel	me-east-215	live	180.306 OMR 1 SKU	2026-09-02	2026-09-27
catalyst-hw307-omani-works-9a1f230f-me-east-215-b-w2a3fb2adc8113d-cho3-48a3-7f53	Elastic Cloud Server	Omantel	me-east-215	live	180.306 OMR 1 SKU	2026-09-02	2026-09-27

1. One row per resource across sources: kind, SKU, region, tags, month-to-date cost.

2. Filter by customer, source, kind or tag; the sum at the top follows the filter.

3. Open a row for the resource's own history (next slide).

Resource detail — one resource, hour by hour



- 1. The usage records behind the number: meter, quantity, window, price applied.
- 2. A resize shows as two SKUs in the same hour; a deleted resource stops at its last window.

Metering — check yourself

- Q1. A server runs 45 minutes of an hour. What quantity does the record carry?
- Q2. Why does a traffic-billed Elastic IP never carry `eip.bandwidth_mbps`?
- Q3. Where does the cost of Keycloak inside a customer's Organization go?

Answers

- A1. 0.75 instance-hours.
- A2. Because the cloud reserved no pipe for it; it is billed per GB from CES traffic instead.
- A3. To the Platform overhead line; it is excluded from the customer's usage.

04

Price books and rating

Where prices, plans, allowances, tiers, discounts and tax are defined — and the order they apply in.

In this module: Price books and scopes · Where plans live · Where allowances and tiers live · Plan vs pay-per-use · Discounts, currency, tax · The waterfall

A price book is the list price

- A book has a scope (cloud or platform), a currency, an annual divisor (annual ÷ 8,760 = the hourly unit price), and items.
- An item is one SKU: unit, unit price, optional terms (tiers or an allowance — next slides).
- A book is assigned per source (module 2). Changing a source's book changes its prices from the next rating run.
- hw307 has four: 'National Cloud 2026 list' (cloud), 'OpenOva plans' (platform, committed), 'Organization PAYG' (platform, pay-per-use), and the partner's derived 'Walk Reseller · retail'.

Price books — the list

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Price books

1 cloud · 3 platform · a book prices one layer, and is assigned to the sources of that layer · coverage measured on the last 30 days of usage

Template CSV

Public catalog preview

New price book

Price books

4

1 cloud · 3 platform

Customers without a source

0

every customer has a cost source

Unpriced SKUs

1

in use in the last 30 days without a rate — cost shows as 0

Scope

All (4)

Cloud (1)

Platform (3)

A cloud book prices cloud SKUs. The two platform books are the two ways an Organization is billed: **committed plans** prices the flat plan line an S / M / L / XL Organization pays, and **pay per use** prices the k8s meters an uncapped flexi Organization pays instead. Neither prices what the other does, so nothing is billed twice.

NAME ↑	SCOPE ↓	CURRENCY ↓	ITEMS ↓	SOURCES ↓	COVERAGE ↓	PUBLIC ↓	STOPPED COMPUTE ↓	EFFECTIVE FROM ↓	
National Cloud 2026 list created 2026-09-06	Cloud	OMR	134	5 4 customers	90 % 1 unpriced	published Withdraw	not billed	always	Open Clone Delete
OpenOva plans committed plans · created 2026-09-07	Platform	OMR	4	4 4 customers	100 %	with the list	billed as running	always	Open Clone Delete
Organization PAYG pay per use · created 2026-09-06	Platform	OMR	3	none	100 % no usage	with the list	billed as running	always	Open Clone Delete
Walk Reseller · retail · OpenOva plans created 2026-09-11	Platform	OMR	4	none	100 % no usage	with the list	billed as running	always	Open Clone Delete

4 rows

- 1. Scope badge (cloud / platform), currency, item count, and how many sources are assigned.
- 2. The derived retail book is read-only: it follows a partner rule (module 9).
- 3. 'New book', 'Import CSV' and 'Clone' start from here.

Catalyst BSS · training · Price books and rating

39

Inside the cloud list book

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Price books / National Cloud 2026 list

National Cloud 2026 list

Cloud book · OMR · annual ÷ 8,760 h · stopped compute not billed · effective always · 134 items · 5 sources of 4 customers

Edit settings

Import CSV

Export CSV

Clone

Delete

Coverage

90 %

of 10 SKUs in use, last 30 days

Unpriced SKUs

1

their usage costs 0 until a rate is added

Items

134

OMR per unit · annual ÷ 8,760

Sources

5

Dhofar Logistics · demo-dhofar-logistics, Gulf Retail Group · demo-gulf-retail, Muscat Health Systems · demo-muscat-health, Omantel · demo-hw307-omani-works-history, Omantel · f27698137bdc4b00ad509cf27f1e5547

SKUs in use by the sources assigned to this book

last 30 days · unpriced first

Assigned sources: Dhofar Logistics demo-dhofar-logistics (Cloud) · Gulf Retail Group demo-gulf-retail (Cloud) · Muscat Health Systems demo-muscat-health (Cloud) · Omantel demo-hw307-omani-works-history (Cloud) · Omantel f27698137bdc4b00ad509cf27f1e5547 (Cloud)

RATE ↑	SKU ↓	UNIT ↓	QUANTITY, 30 D ↓	RESOURCES ↓	UNIT PRICE (OMR) ↓
unpriced	eip.traffic_gb	gb	352.992 gb	12	Add rate
priced	evs.ssd_gb	gb-hour	2,495,471.198 gb-hour	243	0.00022831 OMR
priced	ecs.m7n.2xlarge.8	instance-hour	7,193.438 instance-hour	20	0.29983804 OMR
priced	eip	hour	4,649.805 hour	17	0.02853881 OMR
priced	ecs.m7n.xlarge.8	instance-hour	2,880.783 instance-hour	22	0.15213870 OMR
priced	eip.bandwidth_mbps	mbps-hour	1,868.899 mbps-hour	5	0.00017167 OMR
priced	elb	hour	1,605.792 hour	6	0.01923076 OMR
priced	nat.1	hour	1,605.79 hour	6	0.06037935 OMR
priced	vpc	hour	1,246.972 hour	5	0.00000000 OMR
priced	ecs.s7n.2xlarge.2	instance-hour	0.395 instance-hour	1	0.14621532 OMR

10 rows

1. One row per SKU: ecs.*, evs.*, eip, rds.*, ... with unit and unit price.
2. Coverage tells you which metered SKUs have no price yet ('unpriced') — fix them here, not in the customer.
3. Edit settings: name, currency, annual divisor, public or not (module 1).

WHERE PLANS ARE DEFINED — the 'OpenOva plans' book

■ Chargeback

ANALYSE

● Overview

■ Cost explorer

■ Resources

△ Anomalies

✓ Recommendations

PLAN

■ Capacity

BILL

■ Statements

✕ Collections

○ Budgets

⊗ Reports

FINANCE

📖 Journal

⚖ Reconciliation

📅 Period close

Account mapping

📧 Outbox

CONFIGURE

△ Customers

+ Leads

≡ Partners

§ Contracts

≡ Price books

% Discounts

Price books / OpenOva plans

OpenOva plans

Platform book · OMR · annual + 8,760 h · stopped compute billed as running · effective always · 4 items · 4 sources of 4 customers

Edit settingsImport CSVExport CSVCloneDelete

About this book

committed plans

The committed catalog plans (S 5 / M 9 / L 16 / XL 30 OMR per month for 2 / 4 / 8 / 16 vCPU with 2 GiB per vCPU), billed as one flat plan, <slug> subscription line per Organization per hour: annual = monthly x 12, hourly = annual / 8760 = monthly / 730. The k8s.vcpu / k8s.mem_gb / k8s.pvc_gb meters are deliberately UNPRICED here - under a plan they are the allocation basis, and a bundle has no identifiable per-resource split (M = 2xS, L = 4xS, XL = 8xS), so any per-vCPU rate under this book would be invented. An Organization on the uncapped flexi plan is billed the other way round, per use and with no plan line, by the Organization PAYG book.

Coverage

100 %

of 6 SKUs in use, last 30 days

Unpriced SKUs

0

every SKU in use carries a rate

Items

4

OMR per unit · annual ÷ 8,760

Sources

4

Acme Walk · acmewalk307, Nizwa Fintech · demo-nizwa-fintech, Salalah Tourism Board · demo-salalah-tourism, Sohar Ports Analytics · demo-sohar-ports

SKUs in use by the sources assigned to this book

last 30 days · unpriced first

Assigned sources: Acme Walk acmewalk307 (Platform) · Nizwa Fintech demo-nizwa-fintech (Platform) · Salalah Tourism Board demo-salalah-tourism (Platform) · Sohar Ports Analytics demo-sohar-ports (Platform)

k8s.mem_gb, k8s.pvc_gb, k8s.vcpu are not sold per use: the plan covers them, and they are the allocation basis. No rate is missing.

RATE ↑	SKU ↓	UNIT ↓	QUANTITY, 30 D ↓	RESOURCES ↓	UNIT PRICE (OMR) ↓
basis	k8s.pvc_gb	gb-hour	219,986.82 gb-hour	17	not sold per use
basis	k8s.mem_gb	gib-hour	10,773.426 gib-hour	77	not sold per use
basis	k8s.vcpu	vcpu-hour	4,924.37 vcpu-hour	77	not sold per use
priced	plan.m	plan-hour	569.928 plan-hour	1	0.01232877 OMR
priced	plan.l	plan-hour	84 plan-hour	1	0.02191781 OMR
priced	plan.s	plan-hour	83.954 plan-hour	1	0.00684932 OMR

6 rows

1. A plan is a price-book item like any other: SKU plan.s / plan.m / plan.l / plan.xl, unit plan-hour, resource_kind plan.
2. Unit price = annual ÷ 8,760 = monthly ÷ 730. An Organization on plan M is metered 730 plan-hours a month.
3. What a plan INCLUDES is written as allowances on the plan item (next slide) — that is why there is no separate 'Plans' menu.

The platform's own committed product is a SKU in a platform book, priced per plan-hour; 'Organization PAYG' holds the pay-per-use meters instead.

WHERE ALLOWANCES AND TIERS ARE — 'Add tiers or allowance' on an item

Assigned sources: [Acme Walk acmewalk307 \(Platform\)](#) · [Nizwa Fintech demo-nizwa-fintech \(Platform\)](#) · [Salalah Tourism Board demo-salalah-tourism \(Platform\)](#) · [Sohar Ports Analytics demo-sohar-ports \(Platform\)](#)

Pricing shapes for plan.I

Allowance what the plan includes each billing period

Included quantity (plan-hour) Usage up to this rates to zero; the excess rates at the price below. Leave empty for no allowance.

☐ Carry over Carry the unused part into the next period. An allowance is per billing period and lapses. Carried over, it lasts ONE further period and does not compound.

Volume tiers bands instead of one unit price

Mode

The first band prices the units inside it, the next band the units above it, and so on. The industry default.

No bands: every unit rates at the item's unit price of 0.02191781 OMR.

Leave the last band's upper bound empty: everything above the one before it rates at that price.

What this item charges the same order the bill is rated in

The rest at 0.02191781 OMR per plan-hour.

Price a sample volume (plan-hour) Rated here exactly as the engine would: the allowance first, then the bands.

Sample cost

plan.xl XL plan: 30 OMR/mo plan-hour 360 0.04109589 flat rate 0.04109589 OMR

4 of 4 items 720 OMR/yr listed in total

UNIT PRICE (OMR) ±
not sold per use
not sold per use
not sold per use
0.01232877 OMR
0.02191781 OMR
0.00684932 OMR

and allowances under "Effective price"

1. Each item row has 'Add tiers or allowance' (or 'Edit shapes' once it has one).

2. Allowance: N units of a SKU included per billing period, on the plan item; e.g. plan M includes 50 GB of eip.traffic_gb.

3. Tiers: a price ladder with a mode, graduated or all_units; bands must ascend, the unbounded band last.

4. A contract can add MORE allowance for one customer (module 5); the two add up.

Plan vs pay-per-use — disjoint meters, never both

	OpenOva plans (committed)	Organization PAYG (pay-per-use)
What is metered	plan.s/m/l/xl plan-hours (730 a month)	k8s.vcpu, k8s.mem_gb, k8s.pvc_gb as used
Who is on it	an Organization that bought a size	an Organization on 'flexi'
What the vCPUs cost	nothing extra: the plan covers them (shown as 'not sold per use', not 'unpriced')	the metered rate per vCPU-hour
Allowances	written on the plan item	none by default; a contract may add one
Pinned by	TestPlanAndPAYGBooksAreDisjoint — the two books price disjoint SKU sets	

Plan arithmetic — one Organization, one month

```

plan.m    annual 540.000 → unit 540 / 8760 = 0.061644 per plan-hour
September (720 h, the Organization existed all month)
  plan.m          720 × 0.061644 = 44.383562
  k8s.vcpu        covered by the plan → not sold per use (0.00)
  evs.ssd.gb      200 GB × 720 h × 0.000090 = 12.960000 (metered)
  eip.traffic_gb  180 GB, allowance 50 on plan.m → 130 × 0.012 =
1.560000
-----
list                                     58.903562

```

- The plan is metered by the hour it existed, so a mid-month start bills part of a month without a proration rule.
- The covered meter is reported as 'not sold per use' so nobody chases an 'unpriced' vCPU.
- The allowance is consumed first, the rest rates at the item price.

Volume tiers — the mode is a decision, not a detail

```
ladder    0.010 up to 10,240 · 0.008 up to 102,400 · 0.006 above  
usage     51,200 units
```

```
graduated 10,240 × 0.010 + 40,960 × 0.008 = 430.080000  
all_units 51,200 × 0.008                  = 409.600000
```

```
difference 20.48 on one line — pinned by  
TestGraduatedAndAllUnitsDifferOnTheSameVolume
```

- graduated = Stripe graduated / AWS tiered storage: each band at its own price.
- all_units = Stripe volume / Zuora volume: the whole quantity at the band reached.
- The ladder is validated, never repaired: ascending bounds, unbounded band last, no negative price; a ladder that stops carries its last price upward.

Allowance — included quantity, per period, lapsing

```
plan.m includes    50 GB eip.traffic_gb    (on the plan item)
contract adds     100 GB eip.traffic_gb    (module 5)
included          150 GB per billing period
```

```
September usage 180 GB → 150 free,  $30 \times 0.012 = 0.360000$ 
October   usage 120 GB → 120 free, 0.000000 (30 GB unused lapses)
```

- Allowances add up across plan and contract.
- Nothing carries over unless the contract item says rollover.
- Allowance is applied BEFORE tiers: the included quantity resets the ladder (module 5 shows why that matters).

Discounts — percent or amount, with a combination rule

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Discounts

Negotiated percentages and time-boxed campaigns, applied at statement time on top of list prices

New discount

Combination rule

last changed 2026-09-11 15:01Z

When more than one percent discount applies to the same line, this decides how they combine. Fixed amounts always come off afterwards, whatever the rule. A change applies from the next statement run; an issued statement keeps the rule it was rated under and states it.

Most specific wins

Highest wins

Stack

Compound

Most specific wins.

Per line, the percent discount with the narrowest scope applies: a SKU discount beats a whole-bill one; at the same scope the higher percent wins.

A stackable discount (checkbox in the table below) is added on top of the winner under Most specific wins and Highest wins — a campaign on top of the contract. It changes nothing under Stack or Compound.

Save

EXAMPLE · LIST 100

SKU A (50)

SKU B (50)

DI

Most specific wins	−10	−5
Highest wins	−10	−5
Stack	−15	−5
Compound	−14	−5

10 % on the whole bill and 20 % on SKU A, neither stackable. SKU B only ever sees the 10 %.

Discounts

3

1 for all customers

Active now

1

applied on the next statement run

Scheduled

0

campaign starts later

Ended or off

2

1 ended · 1 switched off

Status

All Active Scheduled Ended Off

Scope

every scope

SCOPE	NAME	KIND	VALUE	APPLIES TO	CAMPAIGN WINDOW	STATUS	ACTIVE	STACKABLE	
All customers	demo: launch campaign 5%	percent off	5 %	whole bill	2026-06-01 → 2026-07-01	ended	☒	☐	Edit Delete
Muscat Health Systems	demo: Muscat Health ECS 10%	percent off	10 %	ecs.m7n.xlarge.8	2026-08-01 → open	active	☒	☐	Edit Delete
Omantel	Q4 launch 15%	percent off	15 %	whole bill	always	inactive	☐	☐	Edit Delete

1. A discount is scoped: all customers, one customer, one SKU or service, a validity window; kind percent or amount.
2. The combination rule on billing settings decides what two matching discounts do: most-specific, highest, stack, or compound.
3. A partner tier is a discount row too (tier_id set, customer NULL) — one engine (module 9).

Currency — one reporting currency, rates with a date

- Books price in their own currency; the Sovereign reports in one reporting currency (billing settings).
- Rates carry an effective date; the ledger stores the converted amount and the rate used, so a later rate change never rewrites history.
- A line that could not be converted is reported as 'unconverted' beside the totals rather than silently dropped.
- Budgets are set in the reporting currency; the explorer can show either.

Tax rules — the kind is not derivable from the rate

kind	rate	what it is	required on the rule
standard	e.g. 5 %	the normal rate for a country, region, category, from a date	—
zero_rated	0 %	taxable at zero — a different line on the return than exempt	the provision
exempt	0 %	outside the tax; a certificate that can expire	the article
reverse_charge	0 % here	the buyer accounts for the tax	the wording
out_of_state	0 %	supply outside the jurisdiction	—

Resolution order: customer exemption → rule for (country, region, category) valid on the date → default. The snapshot is frozen on the invoice at issue and never re-derived.

Tax — rules, categories and the Oman profile

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Tax

What each supply is taxed at, to whom, and from when — by country, region and category of supply, each rule valid over its own dates. An invoice is rated at the rules in force on the day it covers, and prints one summary row per rule that applied.

New rule

Rules in force2every rule on the book

Countries1OM

Categories placed11 category in use

Registered inOMthe seller country every determination turns on

This Sovereign

edit on Billing settings

Registration country OM

Registration number OM1100234567

Legal name OpenOva Sovereign hw307

Address none

Default rate 5 % · what a supply no rule covers is taxed at

Rules

country, then region, then category, then start date

The most specific rule wins: a region rule beats a country-wide one and an exact category beats the catch-all, and only among equally specific rules does the latest start date win. A rule that is not **Standard** charges nothing — its rate is 0 and its **note** is the sentence the invoice must carry.

RULE ↕	APPLIES TO ↕	KIND ↕	RATE ↕	VALID ↓	
Oman VAT standard no note — nothing is printed on the invoice for this rule	OM · every region · every category	Standard	5 %	from 2026-01-01 in force	Edit Delete
Oman VAT zero-rated export Zero-rated export of services.	OM · every region · export	Zero-rated	0 %	from 2026-01-01 in force	Edit Delete

2 rowsDownload CSV

Categories by SKU

Place a SKU

Which category of supply each meter belongs to, so storage and compute can be taxed differently. A row is an exact SKU (k8s.vcpu) or a prefix ending in * (evs.* — the whole storage family); the longest matching prefix wins, and a SKU matched by nothing is the default category, which is the one a rule with an empty category covers.

SKU ↑	CATEGORY ↕	NOTE ↕
-------	------------	--------

1. Rules by country/region/category with validity; the SKU-to-category map tells which rule a line falls under.
2. An issued invoice keeps its own snapshot: changing a rule later changes future invoices only.
3. The e-invoicing seam signs the issued document with the configured key.

The waterfall — one engine, this order, always

#	Step	What it does	Example on 200 units
1	allowance	usage up to the included quantity rates to zero	100 included → 100 to price
2	tiers	the remainder priced by the item's bands or flat price	bands 1.00 ≤100 / 0.10 above → 100.00
3	commitment	the committed head at the committed price, the excess at list	see module 5
4	discounts	customer discounts and a partner tier under the combination rule	−10 %
5	true-up	the shortfall against the monthly minimum, as a named line	see module 5
6	tax	on the net subtotal, true-up included	5 %

rating.ApplyTerms → rating.ApplyDiscounts → rating.TrueUp → rating.TotalsWithDiscount. TestOrderOfOperations pins each step against its plausible alternative.

Price books & rating — check yourself

- Q1. Where do you change what plan L includes?
- Q2. A customer on 'OpenOva plans' shows k8s.vcpu as 'not sold per use'. Is something unpriced?
- Q3. Two discounts match one line: a 10 % for all customers and a 15 % for this customer. What applies under 'most-specific'?
- Q4. Same ladder, 51,200 units: which mode bills more?

Answers

- A1. On the plan.l item in the 'OpenOva plans' book → 'Add tiers or allowance'.
- A2. No. The plan covers it; the meter is the allocation basis, not a missing rate.
- A3. The 15 % (the customer-specific one); 'highest' would also give 15 %, 'stack' 25 %, 'compound' 23.5 %.
- A4. graduated (430.08) bills more than all_units (409.60).

05

Contracts

Negotiated terms for one customer: allowances on top, committed use, a monthly minimum with its true-up, SLA credits and renewal.

In this module: What a contract is · Creating one · Committed-use and allowance lines · Minimum commitment and true-up · SLA credits · Renewal and status · What the statement shows

What a contract is — and is not

- A contract is the set of terms that changes how ONE customer's usage is rated in a period: extra allowances, committed quantities at a rate, a monthly minimum, a term with renewal.
- It is not a document store and not free text: the notes field is the only prose; every other field is arithmetic the rating run reads.
- Only an ACTIVE contract covering the first day of a period rates that period. Draft, expired and cancelled contracts rate nothing.
- Two active contracts on the same day: the one that started last is in force — that is a renegotiation.
- A statement records the contract_id it was rated under, so the invoice says which agreement produced its numbers.

The contract record

Field	Meaning	Example
name	how it is referred to	Nizwa Fintech 2026 agreement
starts_on · term_months · ends_on	the window; the end follows the term (12 months from 2026-01-01 ends 2026-12-31) and can be overridden	2026-01-01 · 12 · 2026-12-31
status	draft · active · expired · cancelled	active
minimum_commitment · currency	the monthly floor on the NET total; empty = none	1,000.00 OMR
auto_renew · renewal_notice_days	renew for another term, and when it appears in the renewals-due list	yes · 60
po_reference	the customer's order for this agreement	PO-4471
items	committed-use and allowance lines (next slide)	3 lines

Contracts — the list and the renewals due

The screenshot displays a software interface for managing contracts. On the left is a dark sidebar with a 'Chargeback' header and a menu organized into sections: ANALYSE (Overview, Cost explorer, Resources, Anomalies, Recommendations), PLAN (Capacity), BILL (Statements, Collections, Budgets, Reports), FINANCE (Journal, Reconciliation, Period close, Account mapping, Outbox), and CONFIGURE (Customers, Leads, Partners, Contracts, Price books, Discounts, Allocation). The 'Contracts' menu item is highlighted.

The main content area is titled 'Contracts' and includes a 'New contract' button. Below the title is a descriptive sentence: 'The agreement a customer's commercial terms hang on: the term, the monthly minimum, and the committed-use and allowance lines the rating engine applies.' Three summary cards are present: 'Active contracts' showing 1 contract, 'Renewals due' showing 0 renewals, and 'Committed monthly' showing 50.000 OMR. Below these is a table titled 'All contracts' with columns for CONTRACT, CUSTOMER, TERM, MONTHLY MINIMUM, STATUS, and RENEWS. A single contract row is shown for 'Acme Walk 2026 agreement'.

CONTRACT	CUSTOMER	TERM	MONTHLY MINIMUM	STATUS	RENEWS
Acme Walk 2026 agreement PO PO-ACME-2026	Acme Walk	12 months · 2026-09-01 → 2027-08-31	50.000 OMR	active	2027-09-01 in 338 days

1. One row per contract with customer, status, term end and minimum.
2. Inside the notice window a contract appears in 'Renewals due' — the list an operator acts on.
3. 'New contract' opens the form (next slide).

Creating a contract

Contracts
The agreement a customer's commercial terms hang on: the term, the monthly minimum, and the committed use and allowance lines the rating engine applies.

New contract

Customer: Acme Walk

Name: e.g. ACME 2026

Starts on: 09/27/2026

Term (months): 12

Monthly minimum: none

Currency: OMR

Status: draft

Renewal notice (days): 30

Purchase order: optional

Auto-renew: ☒ Renew for another term at the end date

Active contracts
1 every contract on the book

All contracts

CONTRACT
Acme Walk 2026 agree PO PO-ACME-2026

1 row

Committed monthly
50.000 OMR
the minimums a period is measured against

MONTHLY MINIMUM **STATUS** **RENEWS**

50,000 OMR	active	2027-09-01 in 338 days
------------	--------	------------------------

1. Customer, name, start, term (end derived), status, monthly minimum, renewal notice, auto-renew, PO.

2. Leave the minimum empty for 'no floor'; a zero is not the same as empty.

3. Save as draft first; set active when signed — only active rates.

The contract page

The screenshot displays the 'Acme Walk 2026 agreement' contract page. The left sidebar contains a navigation menu with sections: ANALYSE (Overview, Cost explorer, Resources, Anomalies, Recommendations), PLAN (Capacity), BILL (Statements, Collections, Budgets, Reports), FINANCE (Journal, Reconciliation, Period close, Account mapping, Outbox), and CONFIGURE (Customers, Leads, Partners, Contracts, Price books, Discounts, Allocation). The 'Contracts' item is highlighted.

The main content area shows the contract details for 'Acme Walk 2026 agreement'. It includes a status bar with 'active', 'Acme Walk', '12 months', '2026-09-01 → 2027-08-31', 'OMR', and 'PO PO-ACME-2026'. Action buttons at the top right are 'Edit', 'Edit lines', 'Issue SLA credit', and 'Delete'.

Key metrics are displayed in four boxes:

- Monthly minimum:** 50.000 OMR. Subtext: a period whose net falls below it carries a true-up line.
- Committed lines:** 0. Subtext: nothing committed.
- Allowance lines:** 0. Subtext: the plan's allowances only.
- Renews:** 2027-09-01. Subtext: 338 days away · notice from 2027-08-01.

The 'Committed use and allowances' section is titled 'No lines' and includes a note: 'A committed-use line prices a quantity of a SKU per period at a negotiated rate, with anything above it at list. An allowance line includes a quantity on top of whatever the plan already includes.' A sub-note states: 'applied before the discounts, by the one rating engine'.

1. KPIs: monthly minimum, committed lines, allowance lines, renews/ends date.

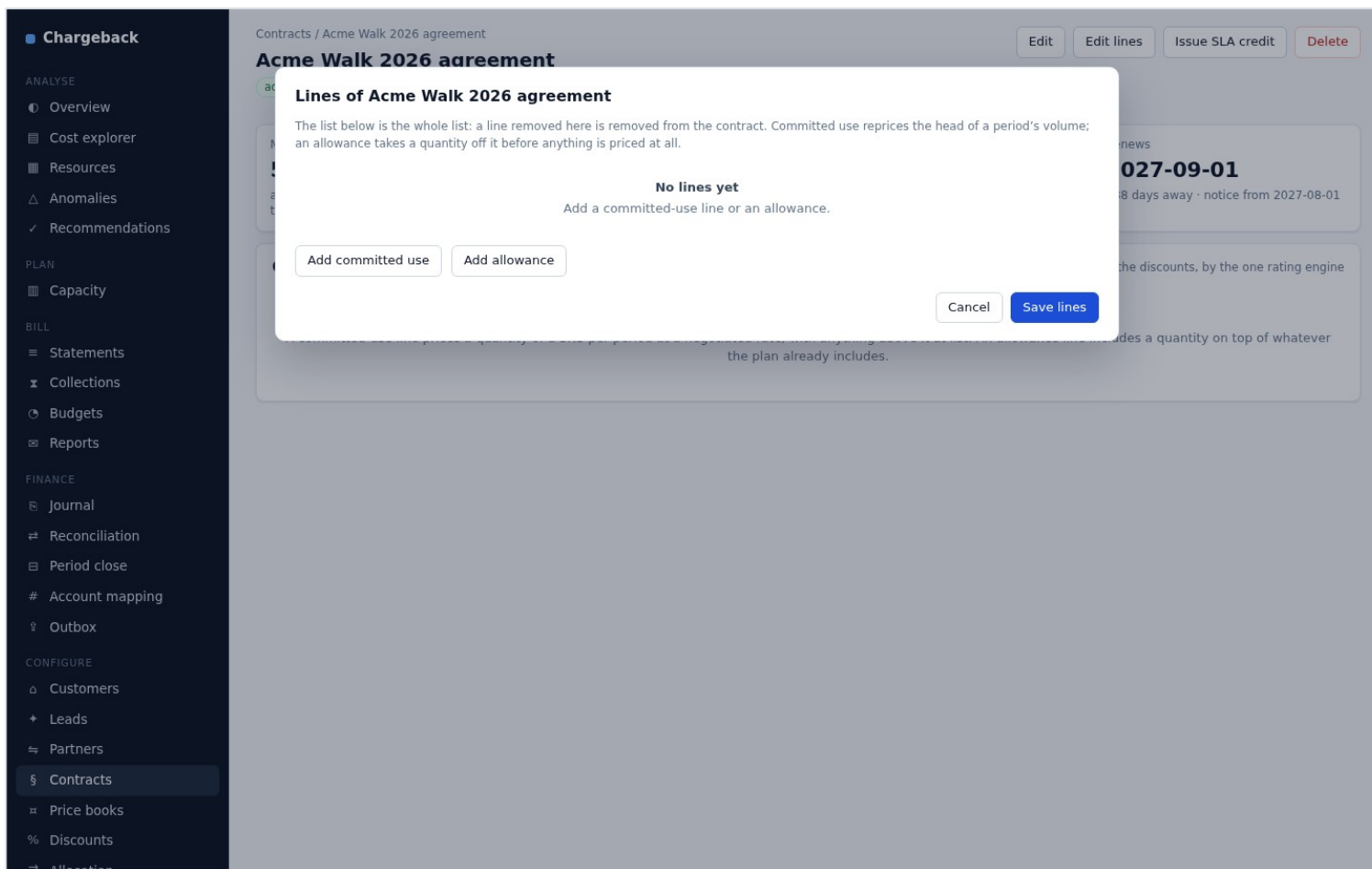
2. 'Committed use and allowances' is the table the rating engine reads; 'Add items' edits it.

3. 'Issue SLA credit' is here because the SLA is a term of this agreement.

The two item kinds

kind	fields	what the rating run does with it
allowance	sku, unit, quantity, rollover	adds quantity to whatever the plan item already includes; usage up to the total rates to zero, per period
commitment	sku, unit, quantity, committed_price OR discount_pct	the committed quantity rates at the committed price (the HEAD of the volume), the excess at list; a commitment at list is refused

Adding committed-use and allowance lines



1. Kind, SKU from the customer's books, quantity per period, and either a committed price or a percentage off list.

2. Rollover on an allowance lets an unused remainder carry into the next period; default is lapse.

Committed use — head of the volume, consumed per period

```
item commitment ecs.m7n.2xlarge.8 7,440 h/period at 0.35 (list
0.50)
    (= 10 servers × 744 h, 12 months, 30 % off)
```

```
month with 5,000 h: 5,000 × 0.35 = 1,750.000000
month with 10,000 h: 7,440 × 0.35 + 2,560 × 0.50 = 3,884.000000
same 10,000 h at list: 10,000 × 0.50 = 5,000.000000
```

```
pinned by TestCommitmentUnderAndOver
```

- Under-consumption is NOT charged by the commitment line — that is what the minimum commitment is for, visibly.
- The commitment displaces the dearest band of a ladder, because it covers baseline usage.
- Each period offers the whole committed quantity again.

Minimum commitment — the true-up is a named line

minimum_commitment 1,000.00 per month (on the NET)

September lines at list	1,100.00	
customer discount 10 %	-110.00	(module 4)
net	990.00	
true-up 1 period	10.00	← added line
subtotal	1,000.00	
tax 5 %	50.00	
total	1,050.00	

- The floor compares against the NET, so a discount agreed in the same contract cannot slide the bill under the minimum.
- The true-up is charged like any service line and taxed; it is explained on the invoice, never hidden in a rate.
- A month above the minimum carries no true-up line at all.

Order of operations — why it is pinned (TestOrderOfOperations)

```
ladder 1.00 up to 100 units, 0.10 above · usage 200

allowance 100, then tiers: 100 free,  $100 \times 1.00 = 100.00$  ← ours
  (allowance 'holding its place' would give  $100 \times 0.10 = 10.00$ )

commitment 100 at 0.50 on the HEAD:  $100 \times 0.50 + 100 \times 0.10 = 60.00$ 
← ours
  (on the tail:  $100 \times 1.00 + 100 \times 0.50 = 150.00$ )

minimum 1,000 · lines 1,100 · discount 200:
  discounts first → net 900 → true-up 100 → 1,000 ← ours
  (true-up first → clears on 1,100, discount → 900, under the
  minimum)
```

- Three places where two defensible readings differ by a factor; the console states ours and the tests refuse the other.

SLA credits — the credit note the agreement promised

- POST /contracts/{id}/sla-credit: pick the statement the breach fell in, enter the percentage owed and the measured availability.
- The amount is that percentage of the statement's total. It is a normal credit note: the same gapless CN-<year>-<seq> number, allocated against the invoice, the remainder becoming account credit, the same ledger entry.
- The credit note records contract_id, sla_pct and measured_availability, so the document says what it answers.
- No parallel 'service credit' table — one ledger to reconcile.

Issuing an SLA credit

The screenshot shows a software interface for managing contracts. A modal dialog titled "Issue an SLA credit" is open, overlaying a background view of a contract for "Acme Walk 2026 agreement". The dialog contains the following fields and text:

- Title:** Issue an SLA credit
- Description:** A credit note against one invoice, numbered and posted to the ledger like any other, recording the percentage owed and the availability actually measured. It reduces what is outstanding first; anything beyond that becomes credit on the account.
- Invoice:** A dropdown menu with the placeholder text "Choose an issued invoice...".
- SLA credit (%):** A text input field with the placeholder "e.g. 10" and a subtext "a percentage of the invoice total".
- Measured availability (%):** A text input field with the placeholder "e.g. 99.2" and a subtext "Recorded on the credit note, so the document says what it answers."
- Reason:** A text input field containing the text "availability 99.2 % against the 99.9 % commitment for August".
- Footer:** Two buttons: "Cancel" and "Issue credit note".

The background interface shows a sidebar with navigation options like "Chargeback", "ANALYSE", "PLAN", "BILL", "FINANCE", and "CONFIGURE". The main content area displays contract details for "Acme Walk 2026 agreement", including a "Monthly minimum" of "50.000 OMR" and a "Renews" date of "2027-09-01".

1. Statement, percentage, measured availability — three fields, one credit note.

2. Example: statement total 1,050.00, SLA owes 10 % → CN of 105.00 against that invoice.

Renewal, expiry, status

- `auto_renew`: at the end date the same agreement moves forward by `term_months`; `renewal_count` increments, `renewed_at` is stamped.
- Without `auto_renew` the contract becomes expired and stops rating the next period.
- `renewal_notice_days` before the end the contract appears in the renewals-due list (`/contracts` and `GET /contracts/renewals`).
- The renew/expire decision runs inside the daily collections pass, idempotently — a restart cannot advance a term twice.
- Cancelled: stops rating from the next period; issued statements keep their `contract_id`.

What the customer sees on the statement

- Lines priced at the committed rate show the rate they were priced at; excess lines show list.
- A true-up line named 'true-up · 1 period' with the shortfall.
- An allowance shows as quantity included, so a 0.00 line is explained.
- The statement carries the contract_id; the customer's Contract tab (module 2) shows the agreement in force.

The customer's Contract tab

The screenshot displays the 'Contract' tab for a customer named 'Gulf Retail Group'. The interface includes a sidebar with navigation options like 'Chargeback', 'ANALYSE', 'PLAN', 'BILL', 'FINANCE', and 'CONFIGURE'. The main content area shows a summary of the customer's status (suspended) and various metrics: Month to date (0 OMR), Forecast month end (needs one complete day), Last month (2026-08) (1.71k OMR), Live resources (0), and Statements (3). Below this is a tabbed interface with 'Contract' selected. The 'Contracts' section shows 'No contract' and a message stating that the customer is rated at its price books alone, with no committed use or monthly minimum. A 'New contract' button is visible in the top right of the 'Contracts' section.

Customers / Gulf Retail Group

Gulf Retail Group suspended

demo-gulf-retail · external · prepaid · Stripe · 1 cloud source · from 2026-06-01 · billing@gulfretail.example

Re-invite Resume Edit Delete

Month to date
0 OMR
vs same days last month (1.49k OMR)

Forecast month end
—
needs one complete day

Last month (2026-08)
1.71k OMR
full calendar month

Live resources
0
1 verified source

Statements
3
0 draft · 3 issued

Overview Account Cost Cost centres Resources Statements (3) **Contract** Discounts Budgets Sources (1) Users (1) Settings Audit

Contracts New contract

No contract

Gulf Retail Group is rated at its price books alone: no committed use, no contract allowance and no monthly minimum. A contract adds the term, those lines and the minimum whose shortfall is invoiced as a true-up.

1. The agreement in force for this customer, its minimum and its lines — read-only here, edited under /contracts.

Contracts — check yourself

- Q1. A contract is 'draft' and the period starts. Does it rate the period?
- Q2. Committed 7,440 h at 0.35; the customer used 6,000 h. Is the unused 1,440 h charged?
- Q3. Net 950 against a minimum of 1,000 with a 5 % tax. What is the total?
- Q4. How is an SLA credit different from a credit note?

Answers

- A1. No. Only an active contract covering the first day rates.
- A2. No. Under-consumption is not charged by the commitment; only a minimum commitment would add a true-up.
- A3. true-up 50 → subtotal 1,000 → tax 50 → 1,050.
- A4. It is not: it IS a credit note with three extra fields (contract, SLA %, measured availability).

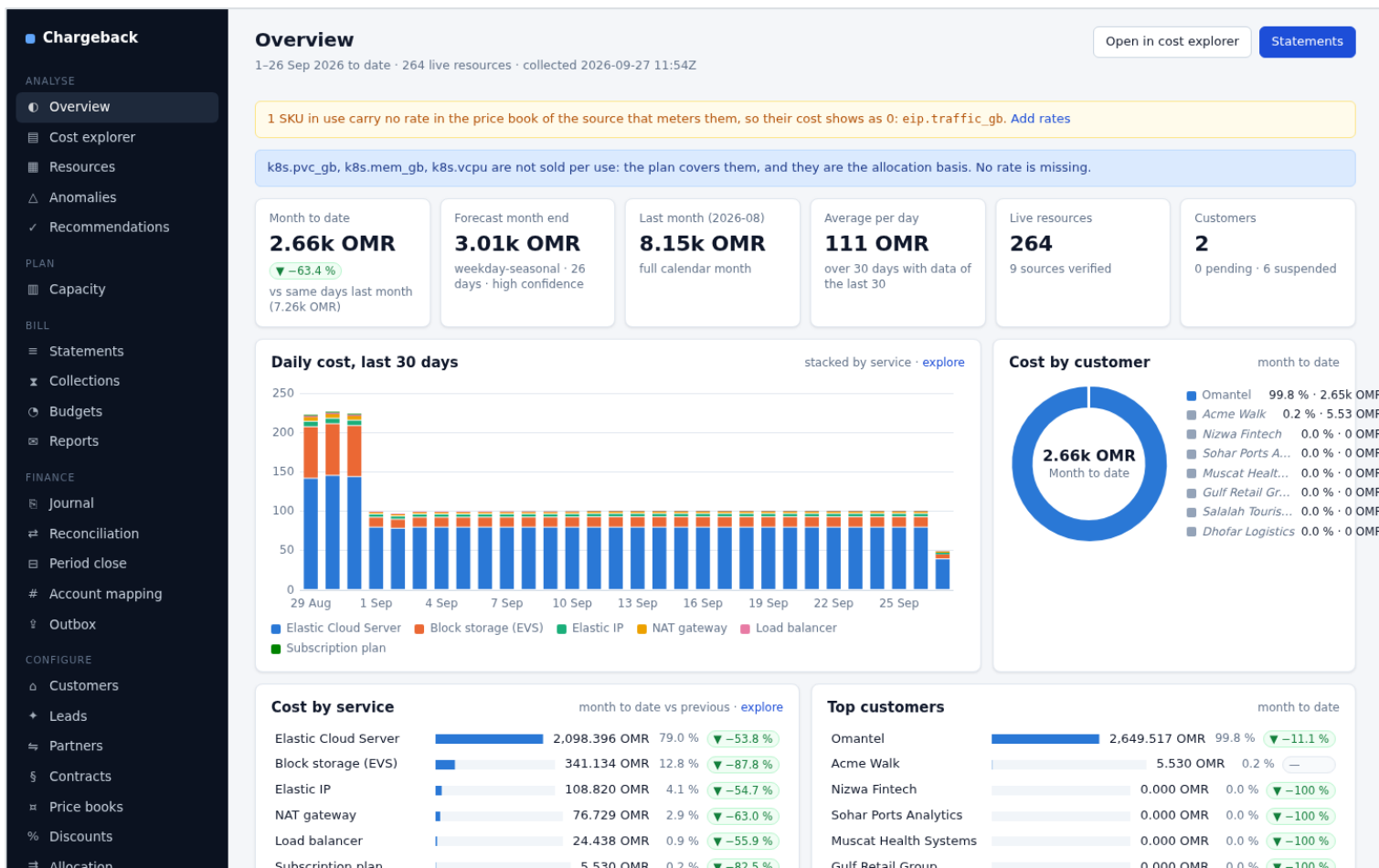
06

Cost analysis

Everyone reads the same priced ledger: overview, explorer, cost centres, allocation, budgets, anomalies, recommendations, reports.

In this module: Overview and the explorer · Cost centres · Allocation is a report · Budgets · Anomalies and recommendations · Reports · The daily rollup

Overview — the month at a glance

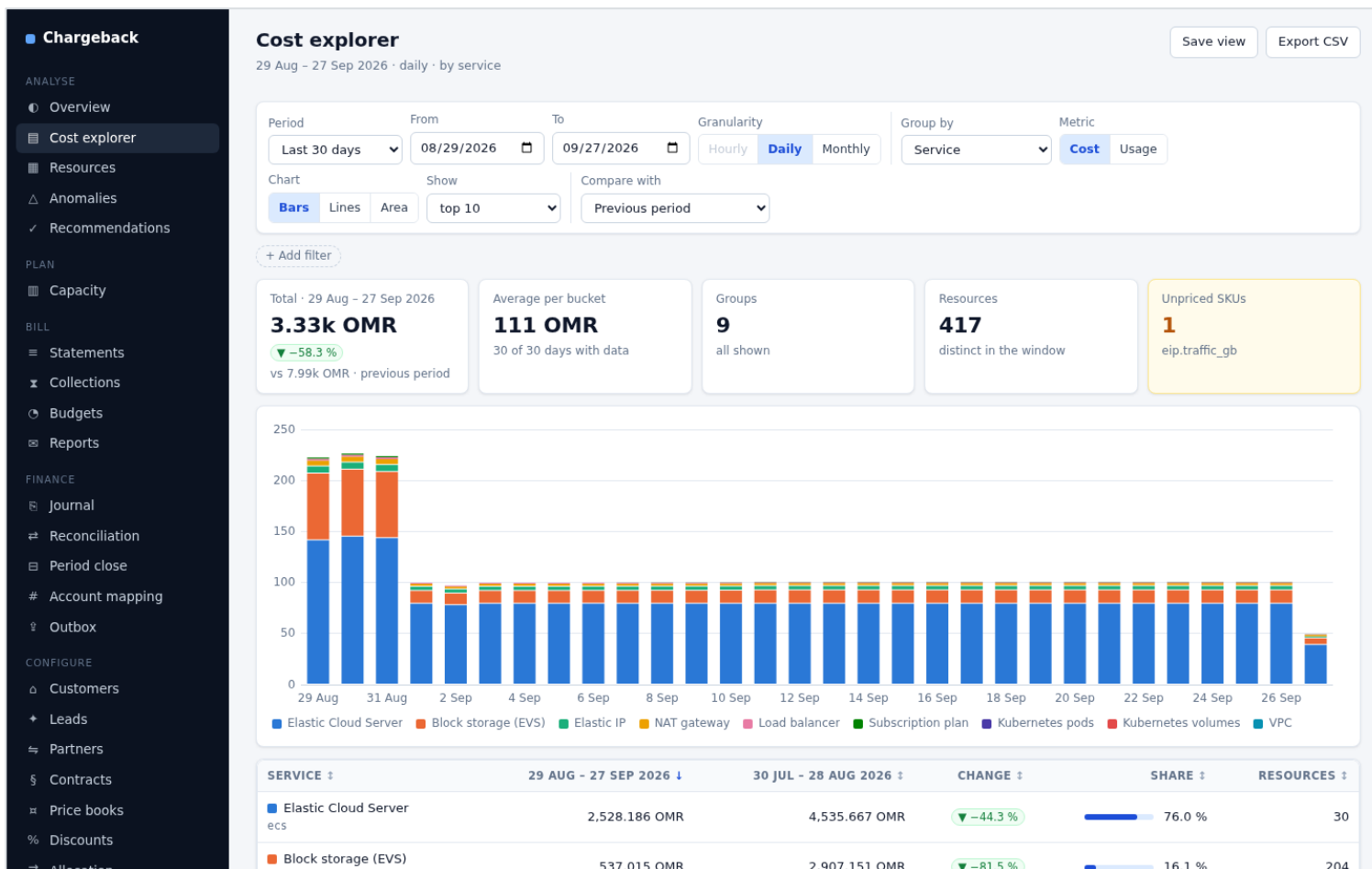


1. Month to date against the previous window; run-rate; top services and top customers.

2. Unpriced and not-sold-per-use are shown beside the total so a gap is visible, never absorbed.

3. Everything here is one read of the daily rollup since 0.1.53 (12 statements, ~0.4 s).

Cost Explorer — group, filter, compare



1. Group by kind, SKU, region, tag, customer, source, cost centre; granularity day / month / hour; top-N with 'other'.

2. Filters include and exclude; a custom compare window sits beside the main one.

3. Export CSV gives exactly the rows on screen.

Cost centres — one customer's spend, labelled

- A cost centre has a code; a rule says 'a resource tagged key = value belongs to it'; an override pins one resource.
- Resolution, in order: resource override → the most specific matching tag rule → the customer's default centre → 'unassigned'.
- It is a dimension like region or tag, so the explorer, the report and the invoice breakdown come for free.
- The invoice breakdown by cost centre sums exactly to the invoice — pinned identity — and is frozen with the closed period.
- Rules read the tags the collector already captures; changing tagging next month changes next month, never history.

Allocation — a report over the two layers, never billing

Chargeback

ANALYSE

- Overview
- Cost explorer
- Resources
- Anomalies
- Recommendations

PLAN

- Capacity

BILL

- Statements
- Collections
- Budgets
- Reports

FINANCE

- Journal
- Reconciliation
- Period close
- Account mapping
- Outbox

CONFIGURE

- Customers
- Leads
- Partners
- Contracts
- Price books
- Discounts
- Allocation

Allocation

A report over the two layers, not billing: the landlord's rated CLOUD cost split across the Organizations by their weighted PLATFORM consumption, against what each is billed

Recalculate

Settings

Basis weights

vCPU-hour: 1
GIB-hour: 1
GB-hour: 1

$$\text{basis} = \text{vCPU-h} \times 1 + \text{GiB-h} \times 1 + \text{GB-h} \times 1$$

this window: $12591.51 \times 1 + 25228.29 \times 1 + 583183.99 \times 1 = 621003.78$

Platform overhead

☒ Keep as a separate line
☐ Distribute across Organizations by share

The Sovereign's own pods and volumes, metered on its internal platform source (it belongs to no customer and is never billed): shown as a row of their own, or folded into every Organization's allocated cost.

Cost pool

☒ The landlord's rated cloud cost for the window
Landlord customer (its cloud sources are the pool)

☐ A manual amount per window

auto: Omantel (the only customer with a verified cloud bill)

The customer whose CLOUD sources carry this Sovereign's cloud bill. The Sovereign itself is not a customer; its own platform footprint is the overhead row below.

Save and recalculate

last saved 2026-09-06 18:18Z

Period: Month to date

From: 09/01/2026

To: 09/27/2026

Cost pool

2.65k OMR

rated cloud cost of Omantel

Allocated

2.65k OMR

100.0% of consumption accounted for

Rated revenue

5.53 OMR

what the Organizations are billed at their price books

Margin

-2.64k OMR

-47,806 % of revenue

ORGANIZATION	TIER	VCPU-H	GIB-H	GB-H	WEIGHTED BASIS
Platform overhead platform	Platform overhead	12032.01	24189.32	558495.69	594717.03
Acme Walk acmewalk307	Organization	559.49	1038.97	24688.29	26286.76
		12591.51	25228.29	583183.99	621003.78

2 rows

Download CSV

Allocated cost by row

1-27 Sep 2026

2.65k OMR allocated

Acme Walk 4.2 % · 112 OMR

Platform... 95.8 % · 2.54k OMR

1. Pool = the landlord customer's rated cloud cost; shares = each Organization's platform usage; overhead = the internal source.

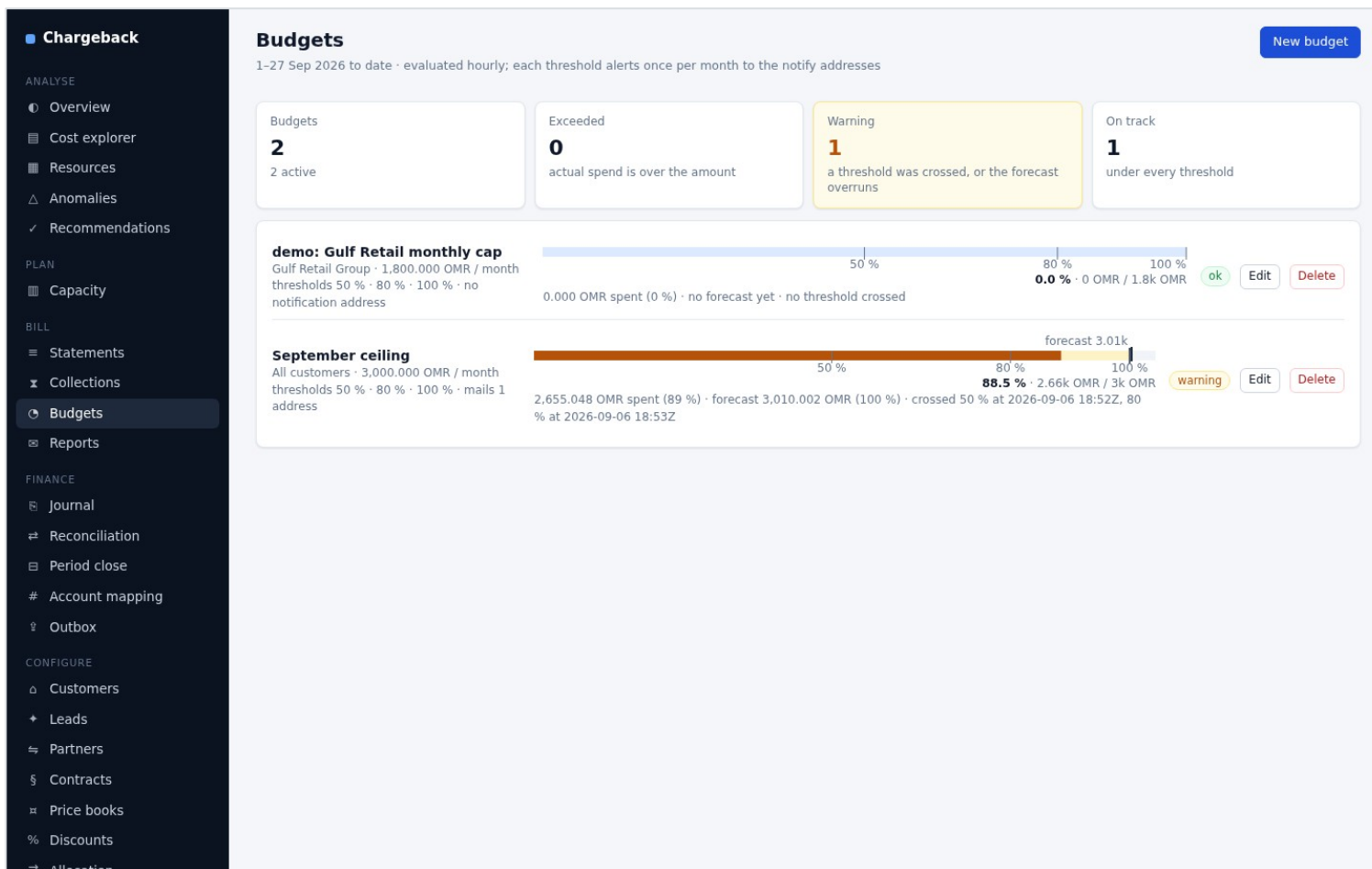
2. The unattributed figure is what no rule explains; the settings editor names the landlord and the basis.

3. Nothing here writes a bill; it explains where the cloud bill went.

Catalyst BSS · training · Cost analysis

74

Budgets — thresholds in the reporting currency

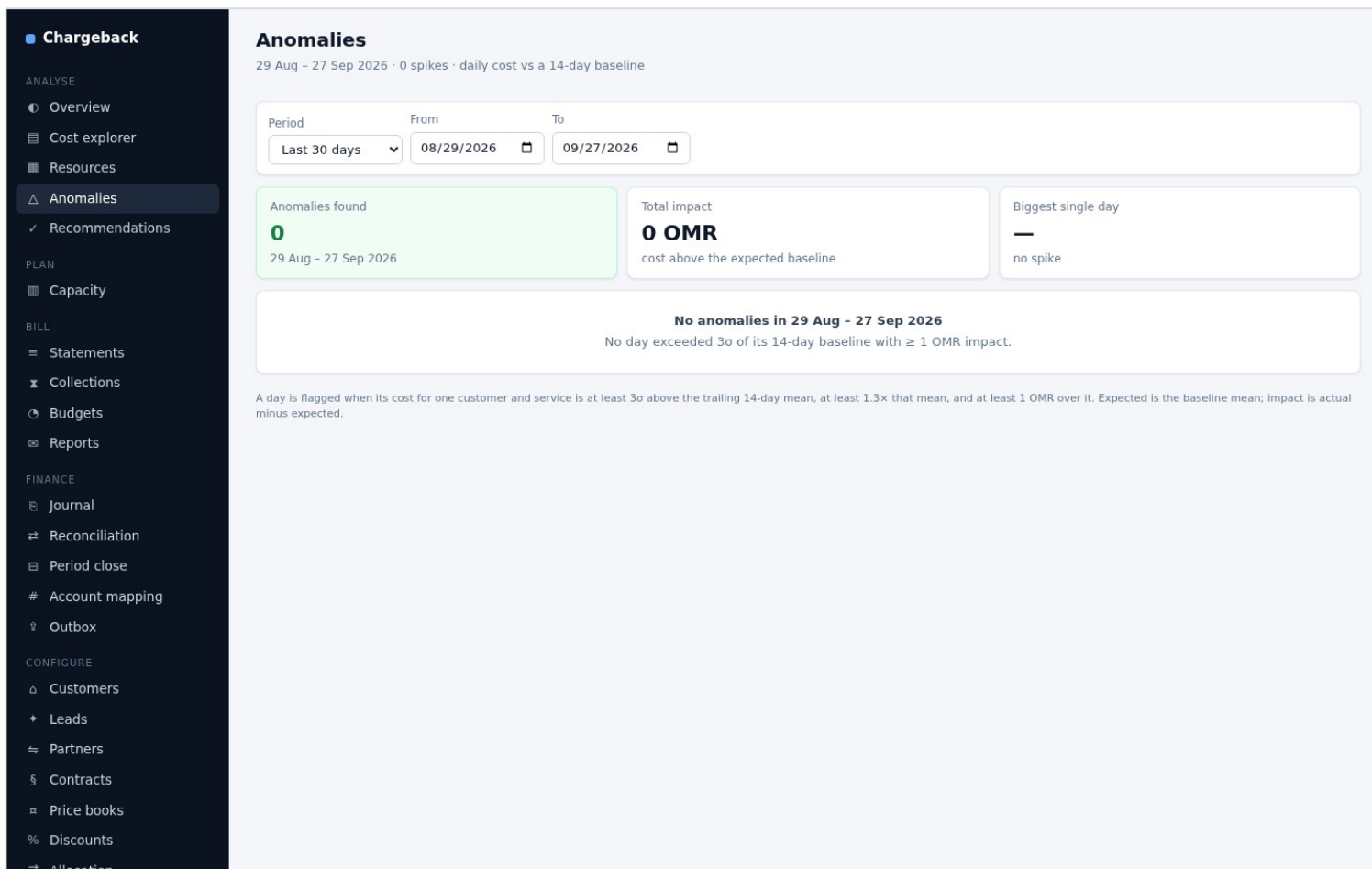


1. Scope all customers or one; a monthly amount; thresholds (e.g. 50 / 80 / 100 %); notification emails.

2. The status bar shows actual (converted), a forecast marker, and which thresholds crossed.

3. The hourly evaluator sends budget.threshold once per crossing (module 11).

Anomalies — the daily series that moved



1. Per customer and per service: today's spend against the expected band from the recent series.

2. An anomaly is a reading, not an alert rule: open the explorer from the row to see what changed.

Recommendations — what to change

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Recommendations

13 findings · 1.08k OMR per month at price-book rates

Potential monthly saving

1.08k OMR

13 recommendations · rate x 730 h

High

2

act this week

Medium

1

worth a look

Low

10

small or uncertain saving

Severity: High 2 Medium 1 Low 10

Type: Low CPU utilisation 10 Customers without a price book 1 Unpriced SKUs 1 Stale cost sources 1

Low CPU utilisation · 10

1,078.205 OMR / month

Seven-day mean CPU under 10 % — the saving is one flavor step down at the price-book rate.

SEVERITY	CUSTOMER	RESOURCE	FINDING	DETAIL	MONTHLY SAVING	ACTION
low	Omantel	catalyst-hw307-omani-works-9a1f230f-me-east-215-b-wd376de 017707d3-c5d3-4016-99df-912d314529f2	Low CPU utilisation	catalyst-hw307-omani-works-9a1f230f-me-east-215-b-wd376de averaged 3.6 % CPU over 167 hourly samples in the last 7 days on m7n.2xlarge.8. Resize one step down to m7n.xlarge.8.	107.821 OMR	Find resource
low	Omantel	catalyst-hw307-omani-works-9a1f230f-me-east-215-b-wbdd113 1a34f644-ec1f-46d0-a694-08d923d0ca69	Low CPU utilisation	catalyst-hw307-omani-works-9a1f230f-me-east-215-b-wbdd113 averaged 7.0 % CPU over 167 hourly samples in the last 7 days on m7n.2xlarge.8. Resize one step down to m7n.xlarge.8.	107.821 OMR	Find resource
low	Omantel	catalyst-hw307-omani-works-9a1f230f-me-east-215-a-we768dc 1fc68d10-9d30-45a0-a0c4-320e0b904e90	Low CPU utilisation	catalyst-hw307-omani-works-9a1f230f-me-east-215-a-we768dc averaged 7.9 % CPU over 167 hourly samples in the last 7 days on m7n.2xlarge.8. Resize one step down to m7n.xlarge.8.	107.821 OMR	Find resource
low	Omantel	catalyst-hw307-omani-works-9a1f230f-me-east-215-a-w9b4787 32aa7cf8-28ad-41ff-a7c8-d2d1423af60c	Low CPU utilisation	catalyst-hw307-omani-works-9a1f230f-me-east-215-a-w9b4787 averaged 0.0 % CPU over 167 hourly samples in the last 7 days on m7n.2xlarge.8. Resize one step down to m7n.xlarge.8.	107.821 OMR	Find resource
low	Omantel	catalyst-hw307-omani-works-9a1f230f-me-east-215-a-w87516b 9dadbe23-b6d4-48e6-afc7-fb4bcd64d49	Low CPU utilisation	catalyst-hw307-omani-works-9a1f230f-me-east-215-a-w87516b averaged 4.4 % CPU over 167 hourly samples in the last 7 days on m7n.2xlarge.8. Resize one step down to m7n.xlarge.8.	107.821 OMR	Find resource
		catalyst-hw307-omani-		catalyst-hw307-omani-works-9a1f230f-me-east-215-b-		

1. Example: the oversized-reservation recommendation compares an Elastic IP's reserved Mbps with its measured traffic and quantifies the saving.

2. Each recommendation names the resource, the evidence window and the money.

Catalyst BSS · training · Cost analysis

77

Reports — scheduled and on demand

Chargeback
ANALYSE
Overview
Cost explorer
Resources
Anomalies
Recommendations
PLAN
Capacity
BILL
Statements
Collections
Budgets
Reports
FINANCE
Journal
Reconciliation
Period close
Account mapping
Outbox
CONFIGURE
Customers
Leads
Partners
Contracts
Price books
Discounts
Allocation

Reports

Scheduled cost reports mailed as plain text: yesterday, the last 7 days or last month, with the sections you pick. Checked every 5 minutes.

Schedules
1
1 active

Sent
3
deliveries in the last 30 days

Failures
1
deliveries that could not be sent

Next due
in 18 h
2026-09-28 06:00Z

REPORT

CADENCE

RECIPIENTS

SECTIONS

NEXT RUN

LAST SENT

hw307 weekly ops report All customers	Weekly on Monday at 06:00 UTC covers the last 7 complete days	emrah.baysal@openova.io	Summary · Top services · Top customers · Budgets · Anomalies · Recommendations	2026-09-28 06:00Z in 18 h	2026-09-21 06:02Z 1 failed in 30 d	Preview	Deliveries	Send now	Edit	Delete
---	---	-------------------------	--	------------------------------	--	---------	------------	----------	------	--------

1 row

1. A saved explorer shape delivered on a schedule to named addresses (notification report.scheduled, five-minute poll).

2. The cost-centre report is here too: one customer's month by centre.

The daily rollup — why every page is fast and still exact

- `cost_usage_daily` holds the priced ledger summed per day, per customer, per source, per SKU, per region, per tags; the hourly chart alone reads live.
- The unit of freshness is a partition (a customer-day). A change to any input re-opens exactly that partition; a reader never touches a stale one.
- Since 0.1.53 every summary (Sovereign, customer, budgets, anomalies) is answered from ONE statement over that table: 98 → 12 statements per overview.
- Exactness is stated, not hoped: batched answers are tested equal to one-by-one answers across a re-opened day, unpriced and unconverted rows, rollup on and off.

Cost analysis — check yourself

- Q1. A resource has an override to centre A and a tag matching a rule for centre B. Where does it land?
- Q2. Does Allocation change any invoice?
- Q3. In which currency is a budget amount entered?

Answers

- A1. Centre A — the override wins over rules.
- A2. No. It is a read-only report over the two layers.
- A3. The reporting currency; the form shows it read-only.

07

Billing — the journey of one period

Run → draft → issue → send → pay → overdue → collect → credit → refund. One customer, real numbers, every screen.

In this module: The lifecycle · Run and the draft · Issue: number, snapshot, document · Send · Payments and part payments · Overdue and collections · Credit notes and refunds · Prepaid and the account · Self-service

The statement lifecycle

From	To	By
(period)	draft	POST /statements/run {period} — the rating run
draft	issued · cancelled	POST /statements/{id}/issue · /cancel; PATCH edits PO and terms while draft
issued	sent · paid · cancelled	POST /statements/{id}/send ({notify:true} also emails the document)
sent	paid · overdue	payments; overdue is DERIVED: sent + due date passed + money outstanding
overdue	paid	payments, or a credit note that clears it
paid	— final	

Our running example: Nizwa Fintech, September 2026

- Profile: billed · postpaid · transfer · PO-4471 · Net 30 · 10 % customer discount · Oman standard 5 %.
- September rating: list 1,000.00 → discount −100.00 → net 900.00 → tax 45.00 → total 945.00.
- We will run, issue, send, receive 600 then 400, watch it go overdue in between, chase it, credit 100 after a dispute, and refund the leftover.

Step 1 — run the period

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Statements

2026-08 · 7 statements in the period

Run period

Period

Customer

Status

August 2026

all customers

All

Drafts

0

nothing awaiting issue

Overdue

0

nothing past its due date

Outstanding

5.52k OMR

issued and sent invoices not yet settled

Total of the selection

5.6k OMR

7 statements · list subtotal – discounts + tax

CUSTOMER ↑	PERIOD ↑	STATUS ↑	DUE ↑	LIST SUBTOTAL ↑	DISCOUNT ↑	TAX ↑	TOTAL ↑	BALANCE ↑	ISSUED ↑	
Dhofar Logistics	Aug 2026	issued INV-2026-00018	2026-10-01 due in 4 days	1,368.892 OMR	—	68.445 OMR	1,437.337 OMR	1,437.337 OMR	2026-09-01 09:00Z	Open CSV
Gulf Retail Group	Aug 2026	issued INV-2026-00014	2026-10-01 due in 4 days	1,963.205 OMR	—	98.160 OMR	2,061.366 OMR	2,000.000 OMR	2026-09-01 09:00Z	Open CSV
Muscat Health Systems	Aug 2026	issued INV-2026-00015	2026-10-01 due in 4 days	2,056.208 OMR	-90.553 OMR	98.283 OMR	2,063.938 OMR	2,063.938 OMR	2026-09-01 09:00Z	Open CSV
Nizwa Fintech	Aug 2026	paid INV-2026-00013	2026-10-01 due in 4 days	14.149 OMR	—	0.707 OMR	14.857 OMR	settled	2026-09-01 09:00Z	Open CSV
Omantel	Aug 2026	cancelled	—	0.000 OMR	—	0.000 OMR	0.000 OMR	—	—	Open CSV
Salalah Tourism Board	Aug 2026	issued INV-2026-00017	2026-10-01 due in 4 days	9.173 OMR	—	0.459 OMR	9.631 OMR	9.631 OMR	2026-09-01 09:00Z	Open CSV
Sohar Ports	Aug 2026	issued	2026-10-01	0.173 OMR	—	0.450 OMR	0.623 OMR	0.623 OMR	2026-09-	Open

- 1. 'Run period' rates every billed customer for the month and writes one DRAFT per customer (a partner's statement follows its customers).
- 2. Running again re-rates drafts only; issued statements never move.
- 3. The list shows status, number, customer, period, total, paid, outstanding.

Step 2 — the draft

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Statements / Sep 2026

Print

Download PDF

CSV

Issue

Delete draft

Nizwa Fintech · Sep 2026

draft

2026-09-01 → 2026-09-30 · OMR · created 2026-09-11 15:01Z · draft, not yet issued

Draft — figures change if the period is run again. Issue it to freeze them.

Invoice

not yet numbered

Edit

Purchase order	none quoted
Payment terms	—
Due	—
Paid	0.000 OMR of 0.000 OMR · settled

From list price to total

OMR

Nothing billed

Every line rated to 0 in this period.

Totals

List subtotal	0.000 OMR
Discounts	+0.000 OMR
Net subtotal	0.000 OMR
Tax 5 %	+0.000 OMR
Total	0.000 OMR

0 rated lines across 0 services · no discount applied · rule: Most specific wins

No rated lines

No usage was collected for Nizwa Fintech in this period, or none of it carries a rate in the price book.

← Statements

- 1. Lines with quantity, unit price, terms applied (allowance, tier, committed rate), discount and tax.
- 2. PO reference and payment terms were copied from the customer; edit them here while draft (PATCH).
- 3. The contract it was rated under is named at the top.

Step 3 — issue: what freezes at that moment

```
POST /statements/{id}/issue
```

```
number      INV-2026-0042    (gapless, per year)
issued_at   2026-10-01
due_at      2026-10-31      (Net 30)
tax_snapshot [{rule: 'Oman standard', kind: standard, rate: 5%,
                        base: 900.00, tax: 45.00}]
document    rendered PDF, the customer's copy
journal     Dr receivable 945 · Dr discounts 100
            Cr revenue.ecs 1,000 · Cr tax_payable 45
status      issued
```

- From here nothing on the statement changes. A correction is a credit note.
- The number is gapless: a cancelled issued statement keeps its number.
- The tax snapshot is the invoice's own; a later rule change does not touch it.

Step 3 — the issued invoice

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Statements / Sep 2026

Omantel · Sep 2026

issued

2026-09-01 → 2026-09-30 · OMR · created 2026-09-11 15:02Z · issued 2026-09-11 15:02Z

Print

Download PDF

CSV

Dispute

Mark sent

Record payment

Credit note

Cancel invoice

Invoice

INV-2026-00020

Edit

Purchase order

none quoted

Payment terms

net 30

Due

2026-10-11 · due in 14 days

Tax

4.37 % on the net subtotal · 46.216 OMR
customer registration OM1100111222 · seller OpenOva Sovereign hw307 OM1100234567

Paid

0.000 OMR of 1,103.366 OMR · 1,103.366 OMR outstanding

Tax summary

2 rules applied · 2 rates

RATE	KIND	CATEGORY	TAXABLE BASE	TAX
5 % Oman VAT standard	Standard	default	924.323 OMR	46.216 OMR
0 % Oman VAT zero-rated export	Zero-rated	export	132.827 OMR	0.000 OMR Zero-rated export of services.
Total			1,057.150 OMR	46.216 OMR

These rows add up to the 46.216 OMR of tax on this invoice.

From list price to total

OMR

The bar chart illustrates the calculation of the final invoice total. It starts with a list subtotal of 1,057.150 OMR, adds a tax of 46.216 OMR, and results in a final total of 1,103.366 OMR. The chart uses blue bars for the subtotal and total, and an orange bar for the tax addition.

Item	Amount (OMR)
List subtotal	1,057.150
Tax	46.216
Total	1,103.366

Totals

List subtotal	1,057.150 OMR
Discounts	+0.000 OMR
Net subtotal	1,057.150 OMR
Tax 4.37 %	+46.216 OMR

1. Number, dates, PO, terms, the frozen tax block, totals, and the document download.
2. Actions from here: Send, Record payment, Credit note, Cancel.

Step 4 — send

- POST /statements/{id}/send marks issued → sent. With {notify: true} it also emails the document (statement.issued).
- The default is not to email, because an operator often marks what was sent by other means (portal, post, an external billing system).
- With the outbox in external mode (module 8) the invoice is also emitted as an event to the billing system of record.
- The customer's users see it under /my/statements immediately, with the PDF.

Step 5 — payments and part payment

```
10 Oct  POST /statements/{id}/payments {amount 600, paid_at, ref
TRF-1187}
        settles 600 of 945 → outstanding 345 · shown as part-paid
        journal  Dr cash 600 / Cr receivable 600

31 Oct  due date passes, 345 outstanding → OVERDUE (derived)

12 Nov  payment 400
        345 settles the invoice → PAID
        55 has nothing to settle → customer_advances (account
credit)
        journal  Dr cash 400 / Cr receivable 345 · Cr
customer_advances 55
```

- A gateway payment goes through the gateway's ConfirmSettlement first; a signed gateway callback books it without an operator.
- Allocate lets you spread one payment over several invoices; Refund returns a whole payment.
- paid_total is the sum of received payments linked to the statement.

Payments on a paid statement

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Statements / Sep 2026

Acme Walk · Sep 2026 paid

2026-09-01 → 2026-09-30 · OMR · created 2026-09-11 13:00Z · issued 2026-09-11 13:01Z

PrintDownload PDFCSVDispute

A dispute on this invoice was resolved: upheld — upheld — a credit note was issued for the disputed amount.

Invoice

INV-2026-00019

Purchase order

none quoted

Payment terms

net 30

Due

2026-10-11 · due in 14 days

Tax

5 % on the net subtotal · 2.500 OMR
customer not tax-registered

Credited

−52.500 OMR · 1 credit note

Paid

0.000 OMR of 52.500 OMR · settled

Credit notes

−52.500 OMR off this invoice · 0.000 OMR can still be credited

CREDIT NOTE	REASON	EFFECT	ISSUED	AMOUNT
CN-2026-00002 partial	dispute upheld; the true-up is billed against a minimum we renegotiated — minimum renegotiated to 0 for September; the true-up is credited	52.500 OMR applied to the invoice	2026-09-11 13:01Z emrah.baysal@openova.io	−52.500 OMR

From list price to total

OMR

Category	Amount (OMR)
List subtotal	50,000
Discounts	0,000
Net subtotal	50,000
Tax 5 %	2,500
Total	52,500

Totals

List subtotal	50.000 OMR
Discounts	+0.000 OMR
Net subtotal	50.000 OMR
Tax 5 %	+2.500 OMR
Total	52.500 OMR

3 rated lines across 2 services · no discount applied · rule: Most specific wins

- 1. The payment history with reference, date, amount and what each settled.
- 2. Allocate and Refund live on the payment row; a refund reverses the whole payment.

Step 6 — collections: aging, reminders, escalation, suspend

- Aging per customer in five buckets (current, 1-30, 31-60, 61-90, 90+), total owed, overdue, oldest due, credit available, suspended badge.
- The schedule lives on billing settings: reminder days as a list (e.g. 3, 10, 20 days after due), escalation day and action.
- The daily evaluator sends collections.reminder per stage and collections.escalation once; every send is in the delivery log.
- Suspend — by escalation, by a prepaid balance at zero, or by the operator — calls the platform: the Organization CR gets spec.suspended with the reason and the actor; Resume clears it.
- Nothing is written off automatically; write-off is an explicit action with its own posting.

Collections — the aging board

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Collections

aging as of 2026-09-27 · 19 open invoices across 7 customers · reminders 3 days before · on the due date · 7, 14, 30 days after · escalate 45 days after, notify

Reminder schedule

Run collections now

Total owed

21,091.518 OMR

19 open invoices

Overdue

9,808.946 OMR

46.5 % of what is owed

Customers overdue

5

of 7 with an open invoice

Suspended

0

held at the platform by this product

CUSTOMER	CURRENT	1-30 DAYS	31-60 DAYS	61-90 DAYS	OVER 90	TOTAL OWED	OVERDUE	OLDEST	
Muscat Health Systems demo-muscat-health · 3 open invoices	2,063.938 OMR	2,208.623 OMR	1,972.264 OMR	—	—	6,244.825 OMR	4,180.888 OMR	58 days c	Invoices Suspend
Gulf Retail Group demo-gulf-retail · 3 open invoices	2,000.000 OMR	1,787.856 OMR	1,376.191 OMR	—	—	5,164.047 OMR	3,164.047 OMR	58 days c	Invoices Suspend
Dhofar Logistics demo-dhofar-logistics · 3 open invoices	1,437.337 OMR	1,726.774 OMR	657.133 OMR	—	—	3,821.244 OMR	2,383.907 OMR	58 days c	Invoices Suspend
Salalah Tourism Board demo-salalah-tourism · 3 open invoices	9.631 OMR	32.104 OMR	29.515 OMR	—	—	71.250 OMR	61.619 OMR	58 days c	Invoices Suspend
Sohar Ports Analytics demo-sohar-ports · 3 open invoices	9.631 OMR	9.631 OMR	8.855 OMR	—	—	28.117 OMR	18.486 OMR	58 days c	Invoices Suspend

1. One row per customer with the five buckets and the actions: Remind now, Escalate, Suspend / Resume.

2. The strip at the top: total owed, overdue, customers overdue, credit available.

Catalyst BSS · training · Billing — the journey of one period

92

Billing settings — numbering, terms, schedule, combination rule

Chargeback

ANALYSE

- Overview
- Cost explorer
- Resources
- Anomalies
- Recommendations

PLAN

- Capacity

BILL

- Statements
- Collections
- Budgets
- Reports

FINANCE

- Journal
- Reconciliation
- Period close
- Account mapping
- Outbox

CONFIGURE

- Customers
- Leads
- Partners
- Contracts
- Price books
- Discounts
- Allocation

Billing

last changed 2026-09-11 15:01Z · invoicing from this Sovereign

Numbering

Invoice numbers are gapless per calendar year and assigned at issue; credit notes are numbered the same way under their own prefix, so one is never mistaken for the other. Changing a prefix affects the next number issued, never an issued document.

Invoice prefix

Credit-note prefix

INV

CN

Upper-case letters, digits or dashes; INV gives INV-2026-00001.

Must differ from the invoice prefix; CN gives CN-2026-00001.

Tax

[rules, categories and rates by country](#)

The Sovereign's own tax identity, frozen onto every invoice at issue (DESIGN.md §9.4), and the default rate a supply no rule covers is taxed at. Rates that differ by country, region or category of supply — and the reverse-charge wording a registered business abroad is invoiced with — are rules, and live on [Configure](#) → Tax.

Tax rate (%)

Tax registration number

5

OM1100234567

Applied to the net subtotal where no rule covers the supply. 0 for no tax.

The Sovereign's own registration, printed on every invoice.

Registration country

Legal name

OM

OpenOva Sovereign hw307

ISO 3166-1 alpha-2. A registered business in another country is invoiced under reverse charge.

The seller as it appears on the invoice.

Address

Collections

[open the aging report](#)

When an invoice is reminded about, measured from its due date, and what happens once it is escalated. The daily pass and Run collections now both follow this (DESIGN.md §9.6).

Reminder days

-3, 0, 7, 14, 30

Reads: 3 days before · on the due date · 7, 14, 30 days after.

Escalate after (days)

Escalation action

15

Notify the operator

1. Invoice and credit-note numbering, default terms, the reporting currency, the discount combination rule, tax defaults.

2. The collections schedule: reminder days, escalation day and action.

3. The billing system of record: internal, or external through the outbox (module 8).

Catalyst BSS · training · Billing — the journey of one period

93

Step 7 — dispute, credit note, refund

Nizwa disputes one line on INV-2026-0042 (already paid).

```
POST /statements/{id}/credit-notes {amount 100, reason}
```

```
  CN-2026-0007 · applied 0 (nothing open) · unapplied 100 → account  
credit
```

```
  journal  Dr credit_notes 100 / Cr customer_advances 100  
  account balance: 55 + 100 = 155 in credit
```

Refund 55 by transfer

```
  journal  Dr customer_advances 55 / Cr cash 55  
  balance 100 – applied to October's invoice at issue
```

- A credit note against an OPEN invoice reduces the receivable first; only the remainder becomes credit.
- The account tab shows the ledger: every payment, credit note, top-up, refund and application in order.

The customer's Account tab — the ledger and the balance

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Customers / Gulf Retail Group

Re-invite

Resume

Edit

Delete

Gulf Retail Group suspended

demo-gulf-retail · external · prepaid · Stripe · 1 cloud source · from 2026-06-01 · billing@gulfretail.example

Month to date
0 OMR
vs same days last month (1.49k OMR)

Forecast month end
—
needs one complete day

Last month (2026-08)
1.71k OMR
full calendar month

Live resources
0
1 verified source

Statements
3
0 draft · 3 issued

Overview **Account** Cost Cost centres Resources Statements 3 Contract Discounts Budgets Sources 1 Users 1 Settings Audit

Balance
5,164.047 OMR
owed by the customer

Credit available
0.000 OMR
nothing on account

Owed
5,164.047 OMR
3 open invoices

Overdue
3,164.047 OMR
past its due date

Account ledger

prepaid wallet · stays up at zero

Top up

Checkout with Stripe

Apply credit

DATE ↕	TYPE ↕	REFERENCE ↕	DEBIT ↕	CREDIT ↕	BALANCE ↕
2026-09-27 08:15Z	Refund	WALK-0151	1.000 OMR	—	5,164.047 OMR owes
2026-09-27 00:00Z	Top-up	WALK-0151	—	1.000 OMR	5,163.047 OMR owes
2026-09-10 12:53Z	Credit note Goodwill credit — demo walk 2026-09-10	INV-2026-00014	—	61.366 OMR	5,164.047 OMR owes
2026-09-01 09:00Z	Invoice	INV-2026-00014	2,061.366 OMR	—	5,225.413 OMR owes
2026-08-01 09:00Z	Invoice	INV-2026-00012	1,787.856 OMR	—	3,164.047 OMR owes
2026-07-01 09:00Z	Invoice	INV-2026-00002	1,376.191 OMR	—	1,376.191 OMR owes

6 rows · credit stays on account until it is applied

Download CSV

1. Balance, then every entry: invoice, payment, credit note, top-up, apply-credit, refund.
2. Top-up and Apply credit are actions here; a prepaid customer's service depends on this balance.

Prepaid — the same machinery, one difference

- The customer tops up first (transfer booked by the operator, or a gateway checkout the customer starts — `account.topup`).
- At issue the invoice settles from the balance immediately; what the balance cannot cover stays outstanding.
- `account.low_balance` warns when the balance falls under the threshold; at zero the wallet suspends the Organization if the setting says so.
- Account credit is universal: a postpaid customer can hold credit too (an overpayment, a credit note).

Statements tab — the customer's own history

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Customers / Gulf Retail Group

Re-invite

Resume

Edit

Delete

Gulf Retail Group

suspended

demo-gulf-retail · external · prepaid · Stripe · 1 cloud source · from 2026-06-01 · billing@gulfretail.example

Month to date

0 OMR

vs same days last month (1.49k OMR)

Forecast month end

—

needs one complete day

Last month (2026-08)

1.71k OMR

full calendar month

Live resources

0

1 verified source

Statements

3

0 draft · 3 issued

Overview

Account

Cost

Cost centres

Resources

Statements 3

Contract

Discounts

Budgets

Sources 1

Users 1

Settings

Audit

PERIOD	STATUS	NET SUBTOTAL	DISCOUNT	TAX	TOTAL	ISSUED	
2026-08-01 → 2026-08-31	issued	1,963.205 OMR	—	98.160 OMR	2,061.366 OMR	2026-09-01 09:00Z	Download CSV
2026-07-01 → 2026-07-31	issued	1,702.720 OMR	—	85.136 OMR	1,787.856 OMR	2026-08-01 09:00Z	Download CSV
2026-06-01 → 2026-06-30	issued	1,310.658 OMR	−68.982 OMR	65.533 OMR	1,376.191 OMR	2026-07-01 09:00Z	Download CSV

1. Every period for this customer with status and outstanding; this is what its own users see at /my/statements.

Self-service — what the customer does without you

- /my/overview and /my/explore: its own cost, same explorer, confined to its customer.
- /my/statements: the list, the PDF download, the payment history.
- /my/account and /my/payment-methods: balance, top-up through the gateway, a saved payment method.
- Dispute an invoice: opens a dispute the operator answers with a credit note or a rejection; an open dispute blocks the period close (module 8).
- /my/users, /my/sources, /my/notifications, /my/cost-centres, /my/budgets: the owner manages its own people, credentials, preferences, centres and budgets.

Billing — check yourself

- Q1. Is 'overdue' a status you can set?
- Q2. A 400 payment arrives against 345 outstanding. Where does the 55 go?
- Q3. Can a customer cancel its own issued invoice?
- Q4. What does {notify:true} on send change?

Answers

- A1. No. It is derived from sent + due date passed + outstanding money.
- A2. To the customer's account as credit (customer_advances).
- A3. No. It can dispute it; the operator answers with a credit note.
- A4. The document is emailed (statement.issued) in addition to the status change.

08

Finance

The journal every billing event posts to, the period close, gateway reconciliation and the seam to an external system of record.

In this module: Accounts · Events → postings · Which month · Reconciliation · Period close · The outbox

The accounts

Key	Default	What posts to it
receivable	1100	what customers owe
cash	1000	money that moved, for a transfer or an internal recharge
gateway_clearing	1010	money a gateway holds between collecting and settling
customer_advances	2100	money held on account: a top-up, a payment beyond what it settled
tax_payable	2200	tax an invoice owes, one line per frozen rule
revenue · revenue.<service>	4000	revenue, per service when the operator added the account
discounts	4800	what discounts took off list, as contra revenue
credit_notes	4900	the reduction of an invoice through a credit note
write_offs · gateway_fees · commission	6100 · 6200 · 6300	a receivable given up · what a gateway kept · what a partner is owed

Events and their postings

Event	Debit	Credit
Invoice issued	receivable (total) · discounts (discount total)	revenue.<service> (list, per service) · tax_payable (per rule)
Payment received	cash or gateway_clearing	receivable (what it settled) · customer_advances (the rest)
Top-up	cash or gateway_clearing	customer_advances
Account credit applied	customer_advances	receivable
Credit note	credit_notes (total)	receivable (applied) · customer_advances (unapplied)
Write-off	write_offs (total)	receivable (applied) · customer_advances (unapplied)
Refund	receivable, or customer_advances for a refunded top-up	cash or gateway_clearing
Settlement fee	gateway_fees	gateway_clearing
Commission statement issued	commission	the partner's account (credit)

Nizwa's September, as the journal sees it

01 Oct	invoice INV-2026-0042 issued		
	Dr receivable	945.00	Cr revenue.ecs 1,000.00
	Dr discounts	100.00	Cr tax_payable 45.00
10 Oct	payment 600 (transfer)		
	Dr cash	600.00	Cr receivable 600.00
12 Nov	payment 400		
	Dr cash	400.00	Cr receivable 345.00
			Cr customer_advances 55.00
	credit note CN-2026-0007 100		
	Dr credit_notes	100.00	Cr customer_advances 100.00
	refund 55		
	Dr customer_advances	55.00	Cr cash 55.00

- Every event balances on its own; the period balance is asserted at close.
- The discount is a contra line, never apportioned into revenue.
- Multi-rate tax posts one tax_payable line per frozen rule.

Which month an event belongs to

- An invoice posts to its billing period. A payment posts to the month it was received, split between what it settled and the advance.
- A credit note posts to the month it was issued, against the invoice it names.
- A closed month accepts nothing new: a late payment against a September invoice posts to the month it arrives.
- GET /finance/journal for a closed period is served from the frozen lines, so an export is stable forever.

The journal

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Journal

every financial event of August 2026 as double-entry lines against the accounts you mapped · closed

Period
August 2026
Download CSV

Total debits
5,748.679 OMR
32 lines

Total credits
5,748.679 OMR

Balance check
0.000 OMR
debits equal credits

Accounts posted to
5
frozen at the close

By account

What a finance system posts for August 2026.

ACCOUNT ↑	NAME ↓	DEBIT ↓	CREDIT ↓	LINES ↓
1100	Trade receivables	5,596.760 OMR	61.366 OMR	7
2200	Tax payable	—	266.512 OMR	6
4000	Revenue	—	5,420.800 OMR	17
4800	Discounts and allowances	90.553 OMR	—	1
4900	Credit notes	61.366 OMR	—	1
		5,748.679 OMR	5,748.679 OMR	

5 rows
Download CSV

Lines

Every line carries the statement, payment or credit note it came from, so any figure traces back.

# ↑	DATE ↓	EVENT ↓	ACCOUNT ↓	DEBIT ↓	CREDIT ↓	CUSTOMER ↓	FROM ↓
1	2026-09-01	Invoice issued	1100 Trade receivables · receivable	14.857 OMR	—	Nizwa Fintech	INV-2026-00013 statement 201d1be7-0e8a-4fa4-ae4f-db46b2aa91b9
2	2026-09-01	Invoice issued	4000 Revenue · revenue	—	14.149 OMR	Nizwa Fintech	INV-2026-00013 statement 201d1be7-0e8a-4fa4-ae4f-db46b2aa91b9
3	2026-09-01	Invoice issued	2200 Tax payable · tax_payable	—	0.707 OMR	Nizwa Fintech	INV-2026-00013 statement 201d1be7-0e8a-4fa4-ae4f-db46b2aa91b9

1. Period selector; one row per posting line with account, debit, credit, the event and the document it came from.
2. Totals per account at the bottom; the balance check is visible before you close.

Gateway settlement reconciliation — four buckets

Bucket	Meaning	What you do
matched	the reference is on both sides for the same amount	nothing
amount-mismatch	both sides have the reference, amounts differ — both figures shown, neither believed	check the gateway fee or a partial capture; post the fee
missing-in-ledger	the gateway settled something no payment records	record the payment or find the failed callback
missing-in-settlement	a payment is recorded the gateway has not settled	wait for the next settlement file or void the payment

Reconciliation

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Reconciliation

a gateway's settlement against what this product recorded — it reports, and you decide

Run a reconciliation

Upload the gateway's settlement file, or fetch it when that gateway publishes a feed. Nothing is corrected either way.

Gateway

From

To

omantel

mm/dd/yyyy

mm/dd/yyyy

Choose File

No file chosen

Fetch from the gateway

The file needs a gateway reference and an amount; a currency, a settled date and a fee are read when they are there.

Runs

No reconciliation has been run yet. Upload a settlement file above.

1. Upload or fetch the settlement, see the four buckets, drill into each row.

2. The settlement fee posts gateway_fees / gateway_clearing from here.

Period close — refuses while anything still moves

- GET /finance/periods/{period} shows, before you press anything: drafts still open, disputes still open, and the balance check as a figure.
- POST .../close refuses with 409 NAMING each draft and each open dispute; refuses with 422 if the journal does not balance.
- Then it stamps the period closed and freezes its journal lines in one transaction.
- After close: no statement in the period can be issued or credited into it, no notification is sent from it, the cost-centre breakdown is frozen.

Periods

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Period close

once a month is closed, no statement in it may be issued, re-run, cancelled or credited

Period
August 2026
Reopen

Status
closed
by emrah.baysal@openova.io

Debits
5,748.679
32 journal lines

Credits
5,748.679

Blocking the close
0
nothing is outstanding

What blocks the close

Nothing. Every statement in August 2026 is decided and no invoice in it is disputed. [Read the journal before closing.](#)

Closed periods

PERIOD ↓	STATUS ↓	CLOSED ↓	LINES ↓	DEBITS ↓	CREDITS ↓	REOPENED ↓
August 2026 2026-08	closed	2026-09-11 emrah.baysal@openova.io	32	5,748.679	5,748.679	—

1 row

- 1. One row per month: open / closed, statements, drafts, disputes, balance check, close button.

Accounts

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Account mapping

the key this product books to, against the code your finance system posts to

Save

Accounts

KEY	ACCOUNT CODE	NAME	LAST CHANGED
receivable what customers owe — an issued invoice debits it, a payment credits it	1100	Trade receivables	2026-09-11 14:49Z migration
cash money that moved, for a transfer or an internal recharge	1000	Cash and bank	2026-09-11 14:49Z migration
gateway_clearing money a gateway holds between collecting and settling	1010	Payment gateway clearing	2026-09-11 14:49Z migration
customer_advances money held on account: a top-up, and a payment beyond what it settled	2100	Customer advances and account credit	2026-09-11 14:49Z migration
tax_payable tax an invoice owes, one line per rule the invoice froze	2200	Tax payable	2026-09-11 14:49Z migration
revenue the catch-all revenue account, for a service with no account of its own	4000	Revenue	2026-09-11 14:49Z migration
discounts what discounts took off the list price, as a contra-revenue line	4800	Discounts and allowances	2026-09-11 14:49Z migration
credit_notes the reduction of an invoice through a credit note	4900	Credit notes	2026-09-11 14:49Z migration
write_offs a receivable given up on	6100	Receivables written off	2026-09-11 14:49Z migration
gateway_fees what a gateway kept, as reconciliation discovered it	6200	Payment gateway fees	2026-09-11 14:49Z migration
commission what a partner is owed on a commission statement	6300	Partner commission	2026-09-11 14:49Z migration

Revenue by service

A service with an account of its own is posted to it; everything else falls back to the catch-all revenue account.
No service has an account of its own yet; all revenue posts to the catch-all account.

Service

Account code

1. The chart of accounts with defaults; add revenue.<service> accounts to split revenue per service.

The outbox — the seam to an external billing system

- Billing settings choose the system of record: internal (BSS issues, collects, posts) or external.
- External: every commercial event (invoice issued, credit note, payment, balance, enforcement) is written to a TMF-shaped outbox and delivered; a CSV exporter serves systems that take files.
- Inbound: signed imports for status, payment, balance and enforcement, so the external system can tell BSS what it did.
- A failed delivery is retried and visible; 'Retry' needs billing.issue. Nothing is dropped silently.
- Adapters for Omantel's gateway and billing wait on Omantel's specification; the seam is what they plug into.

Outbox

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Outbox

invoices are raised and collected here, so what is queued is the journal you export — and what the exporter answered

Export journal

Queued

0

not yet delivered

Failed

0

nothing has been refused

Billing system

this product

invoices are raised and collected here

Documents

Not delivered

All

Nothing is waiting

Every queued document has been delivered.

1. Event, target, attempts, last status, payload preview, Retry.

Finance — check yourself

- Q1. An invoice of 945 with a 100 discount: what is credited to revenue?
- Q2. A September invoice is paid in December; September is closed. Where does the payment post?
- Q3. Two things that stop a period from closing?

Answers

- A1. 1,000 — the list amount; the discount is a separate debit line.
- A2. December.
- A3. An open draft in the period and an open dispute on an invoice in it (and a journal that does not balance).

09

Partners

Resellers and agents: one list price, two independent reductions, a derived margin, and two billing models.

In this module: The waterfall · Resell vs agent · A partner is a party · The derived retail book · Partner scope and portal · Deleting

The waterfall — pinned by TestIntegrationPartnerWaterfallResellAndAgent

```

partner Walk Reseller · tier Silver = 30 % off list
customer Nizwa Fintech · 10 % customer discount
one unit, list 100

list                100.000000
customer discount 10 % -10.000000
CUSTOMER NET        90.000000   what Nizwa pays
tier discount 30 %   -30.000000
PARTNER BUY          70.000000   what Walk Reseller pays us
MARGIN (net – buy)    20.000000   derived, never typed
tax 5 % on 90 = 4.50 → invoice total 94.500000

```

- The two reductions are independent: Nizwa's discount does not move the partner's price and the tier does not move Nizwa's bill.
- There is no markup field anywhere — a second typed price drifts from the first.
- A tier is a discount row (tier_id set, customer NULL); the same combination rule decides two tiers.

Two billing models (partners.bill_to)

	Resell (bill_to = partner)	Agent (bill_to = customer)
Who we invoice	the partner: a wholesale statement, every customer's lines at list less the tier, grouped by end customer	the end customer, at our own books — nothing changes because an agent introduced it
What the end customer sees	showback at the partner's derived retail book; the partner bills them	our invoice, 94.50 in the example
What the partner gets	a real invoice on its account: numbered, due, collected, suspendable	a commission statement: net – buy per end customer (20.00), or commission % × net without a tier; posted as a CREDIT on its account — money we owe, nothing collected
Produced by	the same POST /statements/run, after the customers' statements, reconcilable line for line	the same run

A partner is a party

- Each partner owns one customers row flagged partner (billed postpaid by transfer). Balance, invoices, payments, credit notes, aging and suspension are the customer machinery.
- Its customers are assigned to it; assignment takes effect at the next request and at the next rating run.
- Tiers are the partner's price level: Silver 30 %, Gold 35 %... each a discount row.

The derived retail book (resell only)

```
rule    base = buy | list, ± pct, overridable per service or SKU

base = buy  (70) + 5 %   → 73.500000    no warning
base = list (100) + 5 %  → 105.000000
base = buy  (70) - 10 %  → 63.000000    below the buy price 70 → 'below_buy',
warned, never refused

materialised as a real, read-only price_books row per list book
re-derived on: list change, tier change, rule change, billing-model change,
customer assignment; removed when the partner becomes an agent
```

- Every reader that knows a price book (rating, coverage, export) reads the retail book unchanged.
- Selling at a loss is the partner's decision; hiding it is not ours.

Partners — the list

■ Chargeback

ANALYSE

🔍 Overview

📊 Cost explorer

📄 Resources

⚠️ Anomalies

✓ Recommendations

PLAN

📅 Capacity

BILL

📄 Statements

✖ Collections

🕒 Budgets

📄 Reports

FINANCE

📅 Journal

⚖ Reconciliation

📅 Period close

Account mapping

📧 Outbox

CONFIGURE

△ Customers

+ Leads

≡ Partners

§ Contracts

📖 Price books

% Discounts

🔗 Allocation

Partners

Resellers and agents. One list price per SKU; a tier is a percentage off it, and the margin is derived — never typed.

Partners

1

1 resell · 0 agent

Tiers

1

each a percentage off the list price

Owed by partners

36.908

wholesale statements outstanding

Owed to partners

0.000

commission credited and not yet paid out

New tier

New partner

PARTNER ↑	STATUS ↓	BILLING MODEL ↓	BUY PRICE ↓	CUSTOMERS ↓	RETAIL BOOK ↓	BALANCE ↓
Walk Reseller walk-reseller.ops@walk-reseller.example	active	Resell invoiced the wholesale statement	Gold a percentage off list	1	derived	36.908 owes us

1 row · click a row to open the partner

Tiers

a tier is a set of percent discounts off the list price — the same engine, and the same combination rule, as a customer's discounts

TIER	DISCOUNTS OFF LIST	PARTNERS
Gold 30 % off list	30 % all SKUs	1 Rename Discounts Delete

1. Name, status, model (resell / agent), tier, customers, retail rule, balance.

A partner — tiers, customers, statements, margin, users

Partners / Walk Reseller Suspend Delete

Walk Reseller active
walk-reseller · Resell · Gold · ops@walk-reseller.example

Billing model
Resell
we invoice the partner its wholesale statement; the partner bills its own customers at its retail book

Buy price set by
Gold
a tier is a percentage off the list price

Customers
1
end customers assigned to this partner

Account balance
36.908
owed by the partner

Overview Customers **1** Statements Margin Retail rule Users Account

How this partner is billed

Resell. Each period we invoice **Walk Reseller** a **wholesale statement**: every one of its customers' lines at the Sovereign list price, less its tier (Gold), grouped by end customer. Its customers are priced from its own [retail book](#), derived from its rule — the partner bills them itself.

The margin the partner keeps is what its customers pay less what it pays us. It is derived per line and never typed in: change the list price, the tier or the retail rule, and every figure follows.

Its account, invoices, payments and collections are the ones any party has — the partner owns a customer record of its own (754fffad), so there is one ledger, not two.

1. Tabs: the party's balance and statements, its customers with net / buy / margin per period, the retail rule, and the partner users.

2. Delete is refused by name while customers, tiers, issued statements or ledger entries depend on it; suspend instead.

Partner scope and the partner portal

- A binding at partner:<id> expands, per request, to the partner's customers plus its own party.
- One predicate: every scoped read confines through store.Scope.Confine; a customer named outside the scope is 'not found', never the partner's own rows.
- /partner/overview, /partner/explore, /partner/resources, /partner/statements, /partner/account, /partner/margin, /partner/retail, /partner/users — the same pages, confined.

Partners — check yourself

- Q1. Resell model: who invoices Nizwa?
- Q2. Agent model with no tier and commission 15 %: what is the partner's line on 90 net?
- Q3. Can you type a markup per SKU for a reseller?

Answers

- A1. The partner; we invoice the partner a wholesale statement at list less the tier.
- A2. 13.50 (15 % of the customer net), credited to the partner's account.
- A3. No. A retail rule (base $\pm$ pct) derives the whole book; a retail price below the buy price is warned, not refused.

10

Capacity

What the Sovereign can actually sell: pools of machines, shapes, placements, the classes a pool can enforce, the floor and the envelope.

In this module: Pools and shapes · Placements and classes · Guaranteed floor and burstable envelope · Reclaim · Regions and zones

The model in five lines

- Pool = N identical machines $\times$ a per-machine resource vector (vCPU, memory, storage), in a zone, with reserve machines held back.
- Shape = a sellable SKU's footprint (e.g. `ecs.m7n.2xlarge.8` = 8 vCPU, 64 GB). A family (`ecs.m7n.*`) can stand for all its shapes; exact wins over family.
- Placement = (pool, SKU or family, class): 'this pool sells this shape at this class'. A pool lists the classes it can ENFORCE: guaranteed and spot everywhere; burstable only where the host throttles at runtime.
- A running resource's class: its override $\rightarrow$ its lifecycle tag $\rightarrow$ the most cautious class placed for its SKU.
- Guaranteed floor per resource keeps burstable out of guaranteed room; the overcommit ratio applies only to what is left.

Floor and envelope — the founder's case

```
pool usable          100 vCPU
guaranteed placed (G) 60
guaranteed floor      40
overcommit ratio      2.5   (burstable enforced)
```

```
held      = max(G, floor)          = 60
envelope  = (usable - held) × ratio = 40 × 2.5 = 100
sellable  = held + envelope         = 160
```

```
if G drops to 20: held = max(20, 40) = 40 → envelope 150 → sellable
190
```

- The floor is what guaranteed can always claim back; burstable may never be sold into it.
- Spot may use idle floor and is the first to be reclaimed, newest first.
- Where burstable is not enforced, ratio is forced to 1 and floor to 0 — the arithmetic collapses to plain counting.

Pools

■ Chargeback

ANALYSE

🔍 Overview

📊 Cost explorer

📦 Resources

⚠️ Anomalies

✓ Recommendations

PLAN

📦 Capacity

BILL

📄 Statements

✖ Collections

📅 Budgets

📑 Reports

FINANCE

📖 Journal

⚖ Reconciliation

📅 Period close

Account mapping

📧 Outbox

CONFIGURE

△ Customers

➕ Leads

👤 Partners

§ Contracts

🔢 Price books

% Discounts

🔧 Allocation

Capacity

➕ Add a pool

Pools of machines per availability zone · what each can still sell, by class · when it runs out and when to order · as of 2026-09-27 10:00Z

Pools

1

1 of 1 sized

Pools past 70 %

0

0 critical

Orders owed

0

no pool is growing towards a wall yet

Measured at

2026-09-27 10:00Z

1 cloud source

2 metered SKUs count against nothing: no pool takes them

4 metered SKUs have no shape, so nothing knows what they consume

Open Placements

Open Shapes

Pools 1

Placements 5

Shapes 6

Regions & zones 1

Region

All 1

me-east-215 · Muscat

below 70 %

70 - 85 %

85 % and above

not sized

POOL ↕	ZONE ↕	MACHINES ↕	ENFORCES	BINDS FIRST ↕	SOLD / SELLABLE ↓	ORDER BY ↕	
▶ nodepool walk 2026-09-11: two dual-socket hosts	me-east-215 / me-east-215-a includes usage with no zone	14 96 vCPU · 512 GiB each	Guaranteed Burstable Spot	Memory	9.8 % 704 / 7,167 GiB	Not growing — no order owed	Edit pool Delete

1 row · Open a pool to see why it binds, how many more fit, and what is running on it by class.

Download CSV

1. One row per pool: zone, machines, per-machine vector, classes enforced, ratio, floor, sellable per resource.

2. Add a pool from the row action; edit opens the same modal with its values.

Catalyst BSS · training · Capacity

126

A pool — placements, running resources, reclaim

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Capacity

Pools of machines per availability zone · what each can still sell, by class · when it runs out and when to order · as of 2026-09-27 11:00Z

Add a pool

Pools

1

1 of 1 sized

Pools past 70 %

0

0 critical

Orders owed

0

no pool is growing towards a wall yet

Measured at

2026-09-27 11:00Z

1 cloud source

2 metered SKUs count against nothing: no pool takes them

4 metered SKUs have no shape, so nothing knows what they consume

Open Placements

Open Shapes

Pools 1

Placements 5

Shapes 6

Regions & zones 1

Region

All (1)

me-east-215 · Muscat

below 70 % 70 - 85 % 85 % and above not sized

POOL	ZONE	MACHINES	ENFORCES	BINDS FIRST	SOLD / SELLABLE	ORDER BY	
nodepool walk 2026-09-11: two dual-socket hosts	me-east-215 / me-east-215-a includes usage with no zone	14 96 vCPU · 512 GiB each	Guaranteed Burstable Spot	Memory	9.8 % 704 / 7,167 GiB	Not growing order owed	Edit pool Delete

RESOURCE	USABLE	RATIO	GUARANTEED FLOOR	SELLABLE	BY CLASS	LEFT	TREND
vCPU 96 vCPU each · Reserve 1	1,343 vCPU Raw 1,344	1:1	none	1,343 Sold 88	Guaranteed 80 Burstable 8	1,255 vCPU	Guaranteed not growing Burstable not growing
Memory Binds first · 512 GiB each · Reserve 1	7,167 GiB Raw 7,168	1:1	none	7,167 Sold 704	Guaranteed 640 Burstable 64	6,463 GiB	Guaranteed not growing Burstable not growing

sellable = held + (usable - held) × ratio, where held is the larger of what guaranteed already holds and the guaranteed floor — burstable is only ever sold out of what is left

1. The pool's placements per class, the running resources and their resolved class, fit against the soft wall.

2. Reclaim proposes which spot to give back, newest first; BSS says how much and which, the platform deletes.

Placements

■ Chargeback

ANALYSE

🔍 Overview

📊 Cost explorer

📦 Resources

⚠️ Anomalies

✓ Recommendations

PLAN

📅 Capacity

BILL

📄 Statements

✖ Collections

🕒 Budgets

📑 Reports

FINANCE

📖 Journal

⚖ Reconciliation

📅 Period close

Account mapping

📧 Outbox

CONFIGURE

👤 Customers

➕ Leads

🤝 Partners

💰 Contracts

📈 Price books

📉 Discounts

📊 Allocation

Capacity

Pools of machines per availability zone · what each can still sell, by class · when it runs out and when to order · as of 2026-09-27 10:00Z

Pools

1

1 of 1 sized

Pools past 70 %

0

0 critical

Orders owed

0

no pool is growing towards a wall yet

Measured at

2026-09-27 10:00Z

1 cloud source

2 metered SKUs count against nothing: no pool takes them

4 metered SKUs have no shape, so nothing knows what they consume

Open Placements

Open Shapes

Pools 1

Placements 5

Shapes 6

Regions & zones 1

Region

All 1

me-east-215 · Muscat

Placements

a SKU — or a family of them — sells out of a pool at a class. The same SKU may sit on a pool once per class the pool enforces: three placements, three prices

Place a SKU on a pool

SKU ↑	POOL ↕	ZONE ↕	CLASS ↕	SHAPE / UNIT ↕	RUNNING ↕	
ecs.* a family — nothing under it is metered here now	nodepool	me-east-215 / me-east-215-a	Burstable	each member has its own	0	remove
ecs.c7n.12xlarge.2	nodepool	me-east-215 / me-east-215-a	Burstable	48 vCPU · 96 GiB derived from the name	0	remove
ecs.c7n.12xlarge.4	nodepool	me-east-215 / me-east-215-a	Burstable	48 vCPU · 192 GiB derived from the name	0	remove
ecs.m7n.2xlarge.8	nodepool	me-east-215 / me-east-215-a	Guaranteed	8 vCPU · 64 GiB seed	10 10 resources	remove
ecs.m7n.xlarge.8	nodepool	me-east-215 / me-east-215-a	Burstable	4 vCPU · 32 GiB seed	2 2 resources	remove

5 rows

Download CSV

Metered here, counted against nothing

1. (pool, SKU or family, class) rows; a family shows which SKUs it matched; a class mismatch is flagged.
2. SKU is a dropdown from the price books, two levels: family, then exact shape.

Shapes

■ Chargeback

ANALYSE

🔍 Overview

📊 Cost explorer

📦 Resources

⚠️ Anomalies

✓ Recommendations

PLAN

📦 Capacity

BILL

📄 Statements

🔗 Collections

🕒 Budgets

📑 Reports

FINANCE

📖 Journal

⚖️ Reconciliation

📅 Period close

Account mapping

📧 Outbox

CONFIGURE

△ Customers

➕ Leads

👤 Partners

📄 Contracts

🔗 Price books

% Discounts

🔧 Allocation

Capacity

➕ Add a pool

Pools of machines per availability zone · what each can still sell, by class · when it runs out and when to order · as of 2026-09-27 10:00Z

Pools

1

1 of 1 sized

Pools past 70 %

0

0 critical

Orders owed

0

no pool is growing towards a wall yet

Measured at

2026-09-27 10:00Z

1 cloud source

2 metered SKUs count against nothing: no pool takes them

4 metered SKUs have no shape, so nothing knows what they consume

Open Placements

Open Shapes

Pools 1

Placements 5

Shapes 6

Regions & zones 1

SKUs with no shape

metered in the latest hour, but nothing says how much of each resource one unit consumes

SKU ↕	RUNNING ↓	RESOURCES ↕	REGION ↕	
elb	2 hour	2	me-east-215	➕ Add a shape
nat.1	2 hour	2	me-east-215	➕ Add a shape
vpc	2 hour	2	me-east-215	➕ Add a shape
eip.traffic_gb	0.393 gb	6	me-east-215	➕ Add a shape

4 rows

SKU shapes

how much of each resource one unit of a SKU consumes · the National Cloud list SKUs are seeded, an ECS flavour is read from its name · no per-unit shape on the list: elb, nat.1, vpc

➕ Add a shape

SKU ↑	PER UNIT ↕	SOURCE ↕	
ecs.m7n.2xlarge.8	8 vCPU · 64 GiB	seed	Edit
ecs.m7n.xlarge.8	4 vCPU · 32 GiB	seed	Edit
ecs.s7n.2xlarge.2	8 vCPU · 16 GiB	seed	Edit

1. The footprint per SKU; families expand to their shapes; edit a shape's vector here.

Catalyst BSS · training · Capacity

129

Regions and zones

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Capacity

Pools of machines per availability zone · what each can still sell, by class · when it runs out and when to order · as of 2026-09-27 10:00Z

Pools

1

1 of 1 sized

Pools past 70 %

0

0 critical

Orders owed

0

no pool is growing towards a wall yet

Measured at

2026-09-27 10:00Z

1 cloud source

2 metered SKUs count against nothing: no pool takes them

4 metered SKUs have no shape, so nothing knows what they consume

Open Placements

Open Shapes

Pools 1

Placements 5

Shapes 6

Regions & zones 1

Regions and zones

a region is the code the ledger carries; usage whose zone is unknown lands in the default zone

Add region

REGION CODE ↑	NAME ↓	CLOUD SOURCE KIND ↓	ZONES ↓	POOLS ↓	
me-east-215	Muscat	Huawei project	2	1	Add zoneEditDelete
ZONE CODE NAME POOLS me-east-215-a default zone AZ A nodepool Add a poolEditDelete a a no pools yet Add a poolEditDelete					

1 row

1. A table with row-level add, edit and delete; a zone is added from its region's row.

Capacity — check yourself

- Q1. A pool enforces guaranteed and spot only. What is its overcommit ratio?
- Q2. Usable 100, G 60, floor 40, ratio 2.5: how much is sellable?
- Q3. A resource has no override and no lifecycle tag; its SKU is placed as guaranteed and burstable. What class counts?

Answers

- A1. 1 — ratio and floor exist only where burstable is enforced.
- A2. 160.
- A3. Guaranteed — the most cautious placed class.

11

Access and notifications

Who may do what, at which scope; and what the product sends, to whom, with proof it arrived.

In this module: Permissions and roles · Bindings and directory groups · The notification catalogue · Preferences and channels · The delivery log

The eleven permissions

Permission	Grants
metering.read	every read surface: usage, cost, resources, statements, account, budgets, reports, sources, price books
rating.manage	price books and items, discounts, currency rates
customers.manage	create, edit, invite, import, delete customers and their sources, budgets, schedules
billing.issue	run, issue, edit a draft, send, cancel, delete statements; credit notes; retry the outbox
billing.collect	record, allocate, refund payments; apply credit; run collections; suspend and resume
account.topup	start a gateway checkout for the customer's OWN account — the one money write a customer may make
settings.manage	billing settings, allocation settings, access itself; period close
audit.read	audit trails (Sovereign only)
customer.self.manage	the owner's subset on its own customer: users, source credentials, PO, tax registration
capacity.manage	regions, zones, pools, shapes, placements
sovereign	everything

Access — bindings and directory groups

Chargeback

ANALYSE

Overview

Cost explorer

Resources

Anomalies

Recommendations

PLAN

Capacity

BILL

Statements

Collections

Budgets

Reports

FINANCE

Journal

Reconciliation

Period close

Account mapping

Outbox

CONFIGURE

Customers

Leads

Partners

Contracts

Price books

Discounts

Allocation

Access

9 bindings · 1 configured sovereign-admin · 0 group mappings

Role bindings

who holds which role, at the Sovereign or on one customer

EMAIL ↕	ROLE ↕	SCOPE ↕	GRANTED ↕	
emrah.baysal@openova.io	Sovereign admin	Sovereign	OPERATOR_EMAILS	configured
ops@walk-reseller.example	Partner owner	—	emrah.baysal@openova.io · 2026-09-11 10:45Z	Revoke
accounts@dhofarlogistics.example	Owner	Dhofar Logistics	migration · 2026-09-10 16:59Z	Revoke
billing@gulfretail.example	Owner	Gulf Retail Group	migration · 2026-09-10 16:59Z	Revoke
digital@salahahtourism.example	Owner	Salalah Tourism Board	migration · 2026-09-10 16:59Z	Revoke
emrah.baysal@openova.io	Owner	Acme Walk	migration · 2026-09-10 16:59Z	Revoke
emrah.baysal@openova.io	Owner	Omantel	migration · 2026-09-10 16:59Z	Revoke
finance@muscathealth.example	Owner	Muscat Health Systems	migration · 2026-09-10 16:59Z	Revoke
it@soharports.example	Owner	Sohar Ports Analytics	migration · 2026-09-10 16:59Z	Revoke
ops@nizwafintech.example	Owner	Nizwa Fintech	migration · 2026-09-10 16:59Z	Revoke

10 rows

Email

name@example.com

Role

Finance viewer

Sovereign-wide

Grant

Directory group mappings

read from the X-Forwarded-Groups header the SSO gate forwards

No group mapped

1. Role bindings: email → role → scope (sovereign, customer:<id>, partner:<id>).

2. Directory group mappings: a group from the SSO gate → role → scope; the permission-gated console follows.

3. /api/v1/me tells any page what the session may do; a missing permission hides the action and 403s the route.

The notification catalogue

Event	What it is	Mandatory	Sent by
auth.pin	the one-time sign-in code	yes	POST /auth/pin/request
customer.invite	the activation link	yes	POST /customers/{id}/invite
statement.issued	the invoice	yes	issue, and send with notify
collections.reminder	one dunning stage	yes	the daily collections pass
collections.escalation	the overdue escalation	yes	the daily collections pass
account.low_balance	the prepaid low-balance warning	no	after any account change
budget.threshold	a budget threshold crossing	no	the hourly budget pass
report.scheduled	a scheduled cost report	no	the five-minute report poll

Preferences, channels, templates, the log

- Preference: per customer and recipient, an event can be switched off unless it is mandatory; the effective state is shown per event.
- Channels: one interface; email has a real transport (the Sovereign's SMTP); a channel declared without a transport answers 'unavailable' honestly instead of pretending.
- Templates per event and locale; the subject and body are golden-tested to render exactly what the old inline renderer did.
- Delivery log: every attempt with recipient, channel, status — sent, retrying, failed (the visible one), suppressed, unavailable — and the reason.
- Nothing is sent from a closed period.

Notifications — catalogue, preferences, delivery log

■ Chargeback

ANALYSE

● Overview

■ Cost explorer

■ Resources

△ Anomalies

✓ Recommendations

PLAN

■ Capacity

BILL

■ Statements

✕ Collections

○ Budgets

⊞ Reports

FINANCE

📖 Journal

⚖ Reconciliation

📅 Period close

Account mapping

📧 Outbox

CONFIGURE

△ Customers

+ Leads

≡ Partners

§ Contracts

≡ Price books

% Discounts

≡ Allocation

Notifications

Every message this product can send, who receives it, and what happened to each one. A notice the operator marks mandatory — an invoice, a dunning reminder — cannot be switched off by anyone.

Events

8

5 mandatory · 3 a recipient may switch off

Channels

2

1 declared without a transport

Delivered

1

attempts accepted by a channel in the last 7 days

Not delivered

0

nothing failed in the last 7 days

sms is declared and cannot deliver. SMS has no transport in this build: Omantel's gateway specification has not been provided, so there is no endpoint, credential shape, message payload or delivery-receipt semantics to implement against. The channel is declared so that preferences, the console and the delivery log already carry it; it refuses every send and records the refusal rather than pretending to deliver. Wiring it is one Channel implementation registered here, once the specification exists.

Events

what the product sends, and the template it sends it from

EVENT ↕	GROUP ↑	SWITCHABLE ↕	CHANNELS ↕	SOVEREIGN DEFAULT ↕	SUBJECT	
Sign-in code auth.pin	Access	Mandatory cannot be switched off	email	On no preference set	Example Your sign-in code an example — nothing has been sent yet	Edit Reset
Account activation customer.invite	Access	Mandatory cannot be switched off	email	On no preference set	Example Activate your chargeback account an example — nothing has been sent yet	Edit Reset
Statement issued statement.issued	Billing	Mandatory cannot be switched off	email	On no preference set	Statement for Omantel — June 2026: 1951.362 O last sent 2026-09-11 21:25Z	Edit Reset
Payment reminder collections.reminder	Collections	Mandatory cannot be switched off	email	On no preference set	Overdue: Invoice INV-2026-00012 for 1787.85612 last sent 2026-09-14 14:34	Edit Reset
Overdue escalation collections.escalation	Collections	Mandatory cannot be switched off	email	On no preference set	Action required: invoice INV-2026-00006 for 8.854 last sent 2026-09-14 14:33Z	Edit Reset
Low balance account.low_balance	Collections	a recipient may switch it off	email	On no preference set	Example Low balance: 12.5 OMR left on your ac an example — nothing has been sent yet	Edit Reset
Budget threshold budget.threshold	Cost	a recipient may switch it off	email	On no preference set	Example Budget Acme cap: 80% of 100 OMR re an example — nothing has been sent	Edit Reset
Scheduled cost report report.scheduled	Cost	a recipient may switch it off	email	On no preference set	Cost report: hw307 weekly ops report — 14-20 Se last sent 2026-09-21 06:02Z	Edit Reset

8 rows

Download CSV

1. Three tabs: the catalogue with effective channels per event, the preference rules, and the log with status and reason per attempt.

Access & notifications — check yourself

- Q1. Which single permission lets a customer move money?
- Q2. Can a customer switch off the invoice email?
- Q3. A delivery shows 'unavailable'. Is the address wrong?

Answers

- A1. account.topup — and only for its own account, through the gateway.
- A2. No. statement.issued is mandatory; low-balance, budget and report mails can be switched off.
- A3. No. The channel has no transport configured; 'failed' is the one that means the send was attempted and lost.

12

Putting it together

The journey once more, the vocabulary, and the exercises to run on hw307.

The journey, with the numbers from this course

Step	Nizwa Fintech example	Page
Lead	priced 10 × ecs.m7n.2xlarge.8, left an address	/estimate → /leads
Customer	billed · postpaid · transfer · PO-4471 · Net 30	/customers/new
Sources	cloud project on 'National Cloud 2026 list'; Organization on 'OpenOva plans'	?tab=sources
Metering	usage records per hour; resize split; EIP by charge mode	/resources
Rating	list 1,000 → discount 10 % → net 900 → tax 45	/pricebooks · /discounts · /tax
Contract	7,440 h committed at 0.35; minimum 1,000 → true-up when short	/contracts
Billing	INV-2026-0042 · 600 + 400 paid · 55 advance · CN 100 · refund 55	/statements · /collections
Finance	postings balance; September closed; December payment posts to December	/finance/*
Partner	Walk Reseller buys at 70, Nizwa pays 90, margin 20	/partners

Vocabulary — one line each

Commercial

- Customer — the billed party. Partner — a party that resells or introduces.
- Source — where usage comes from; priced by one book.
- Price book — list prices by scope. Plan — a plan-hour item in a platform book.
- Allowance — included quantity per period. Tier — a price ladder with a mode.
- Contract — negotiated terms: allowances, commitments, minimum, SLA, renewal.
- Discount — percent or amount under a combination rule; a tier is one.

Operational

- Usage record — one resource, one meter, one hour. Priced ledger — usage × price.
- Statement — a customer's period: draft → issued → sent → paid; overdue is derived.
- Credit note — the only way to correct an issued invoice. Advance — money on account.
- Journal — every event's postings; a closed period is frozen.
- Outbox — the seam to an external billing system.
- Pool · shape · placement · class · floor — what can be sold, and how much.

Exercises on hw307 (do them in this order)

- 1. On /estimate price 2 servers and a disk, save with an address, find it on /leads.
- 2. Create customer 'Training Co' (billed, postpaid, transfer, Net 30); add a platform source on 'OpenOva plans'.
- 3. In 'OpenOva plans' open plan.m → Add tiers or allowance → 50 GB eip.traffic_gb. Explain where the 51st GB is priced.
- 4. Create a contract for Training Co: minimum 500, one allowance of 100 GB, one commitment; set it active.
- 5. Run September, open the draft, edit the PO, issue, send with notify, record a part payment, check the account tab.
- 6. On /collections send a reminder; suspend; resume. Find the three sends in /notifications → log.
- 7. Issue a credit note of 50 against the invoice; read /finance/journal for the month; try to close the period and read the refusal.
- 8. Make 'Walk Reseller' the partner of Training Co; run again; compare the wholesale statement with Training Co's.
- 9. On /capacity add a pool with burstable enforced, ratio 2, floor 30; place ecs.m7n.* as guaranteed and burstable; read sellable.

Questions, per module

Ask by module number and screen; every slide names its page path.

Catalyst BSS · products/chargeback · DESIGN.md is the reference behind every slide.